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ABC-based forecasting in misspecified state space models☆

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ABSTRACT

Approximate Bayesian Computation (ABC) has gained popularity as a method for conducting inference and forecasting in complex models, most notably those which are intractable in some sense. In this paper, we use ABC to produce probabilistic forecasts in state space models (SSMs). Whilst ABC-based forecasting in correctly-specified SSMs has been studied, the misspecified case has not been investigated. It is this case that we emphasize. We invoke recent principles of 'focused' Bayesian prediction, whereby Bayesian updates are driven by a scoring rule that rewards predictive accuracy; the aim being to produce predictives that perform well in that rule, despite misspecification. Two methods are investigated for producing the focused predictions. In a simulation setting, 'coherent' predictions are in evidence for both methods. That is, the predictive constructed using a particular scoring rule often predicts best according to that rule. Importantly, both focused methods typically produce more accurate forecasts than an exact but misspecified predictive, in particular when the degree of misspecification is marked. An empirical application to a truly intractable SSM completes the paper.

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1. Introduction

State space models (SSMs) are a key forecasting tool used in a wide range of areas, including astronomy, ecology, economics and finance. Whilst the predictive distributions for such models are often readily accessible via Bayesian Markov chain Monte Carlo (MCMC) methods (see Fearnhead, 2011, and Giordani et al., 2011, for reviews) or particle MCMC variants thereof (Andrieu et al., 2011; Flury & Shephard, 2011), challenges remain when the model is 'intractable' in some sense. Intractable SSMs come in two varieties: *First*, those for which the dimension of either the data or the latent states, or both, is very large; *Second*, those for which some component of the model does not admit an analytical representation as a probability mass, or density function. Exact Bayesian methods based on (P)MCMC are computationally impracticable in the first case and typically infeasible in the second.

High-dimensional SSMs are increasingly tackled via variational methods (Blei et al., 2017; Zhang et al., 2018), which are able to produce approximations of both the posterior and predictive distributions in a reasonable computing time (Tran et al., 2017; Chan & Yu, 2022; Loaiza-Maya et al., 2022; Frazier et al., 2023; Koop & Korobilis, 2023; Quiroz et al., 2023). SSMs with unavailable components on the other hand, have been managed via approximate Bayesian computation (ABC) (Dean et al., 2014; Creel & Kristensen, 2015; Frazier et al., 2019; Martin et al., 2019). This paper continues in the latter vein by looking explicitly at the use of ABC to conduct forecasting in SSMs characterized by this second form of intractability.

Whilst ABC-based forecasts for SSMs were first systematically studied in Frazier et al. (2019), the emphasis therein was entirely on the correctly specified case; with exploration of the performance of the forecasts under

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misspecification of the data generating process (DGP) left as a future goal. It is that goal which we now pursue as part of this paper. Moreover, for the first time, we meld the principles of ABC-based forecasting with recent developments in so-called 'generalized', 'loss-based', or 'focused' Bayesian prediction, in which performance according to a user-specified measure of predictive accuracy is the goal, rather than correct model specification *per se* (Loaiza-Maya et al., 2021; Frazier et al., 2024). In so doing, we show that not only does ABC provide a way of managing *intractable* models (*i.e.*, the usual motivation for ABC), it is also a feasible way of implementing loss-based prediction *per se* in the state-space setting, due to the computational difficulty of specifying 'exactly' a loss-based criterion in an SSM.

ABC constructs an approximation to an exact posterior via simulation from the assumed model, avoiding specification of the (unavailable) likelihood function altogether. In the simplest scenario, draws of the unknown parameters from the prior that yield simulated data that 'match' the observed data – through the prism of selected summary statistics – are used to produce the approximation to the exact posterior; the accuracy of the approximation depending, amongst other things, on the closeness of the summaries to sufficiency. Under the condition that the assumed model correctly specifies the true DGP, and under appropriate regularity, the ABC posterior has been shown to be Bayesian consistent for the true parameter (vector), to be asymptotically normal, and to produce a posterior mean with a sampling distribution that is asymptotically normal (Frazier et al., 2018). In an explicitly state space setting Martin et al. (2019) also prove the consistency of the ABC posterior for the static (or 'global') parameters. Moreover, Frazier et al. (2019) have shown that – again, under correct specification – ABC-based forecasting (which they term 'approximate Bayesian forecasting', or ABF) yields equivalent results to exact Bayesian forecasting in large samples, with both the approximate and exact predictives 'merging', in turn, with the true predictive distribution. Once again, this result nests the case of prediction in an SSM.

Critically, as emphasized, all the ABC work cited above operates explicitly in a correctly-specified scenario. In contrast, Frazier et al. (2020) explore the properties of ABC inference under misspecification, and demonstrate that while the ABC posterior does concentrate onto an appropriately defined pseudo-true value in the limit, it possesses nonstandard asymptotic behaviour. The implications of these results for ABC-based forecasting are left unexplored – and that is the key focus of the current paper, specifically in the context of misspecified *SSMs*. However, once the assumption of the correct model specification is no longer maintained, the whole tenor of the Bayesian forecasting exercise needs to change to one in which focus is directed at the particular goal of the forecasting exercise. Such is the argument put forward in the recent forecasting work by Loaiza-Maya et al. (2021) and Frazier et al. (2024), and is the one that we pursue herein.

In short, the aim is to replace the (potentially misspecified) likelihood function in the conventional Bayesian update with a criterion function that rewards a user-specified form of forecast accuracy or, equivalently, penalizes the associated forecast loss. In the current context, in which inference is conducted via ABC, the approach involves selecting summary statistics that give weight to ABC draws, which will – in turn – lead to predictive accuracy in a selected scoring rule. Accordingly, we refer to this process as loss-based approximate Bayesian forecasting, or 'loss-based ABF' for short. We implement this approach by embedding within the ABC algorithm a loss-based criterion function defined in terms of an 'auxiliary model' (Drovandi et al., 2011; Drovandi et al., 2015; Martin et al., 2019) that is a good match to the assumed SSM, but which has a closed-form predictive density. The sample criterion is built from the scoring rule of interest, with maximization of that criterion producing the summary statistics used to drive the ABC algorithm. We compare this approach with the alternative of using the auxiliary model *itself* (*i.e.* independent of the ABC mechanism) to produce suitably focused predictive distributions via a generalized Bayesian update.

The paper proceeds as follows. Section 2 provides the appropriate background for the forecasting approach advocated herein by outlining the key aspects of performing exact and ABC-based forecasting in *correctly*-specified SSMs. In Section 3, we then illustrate the proposed loss-based ABF approach in a misspecified setting. As a comparator, as noted above, the auxiliary model adopted within the ABC algorithm is used, in its own right, to produce a loss-based predictive *directly* via a generalized Bayesian update (without an ABC step). To keep the nomenclature clear, we refer to this second approach as focused Bayesian prediction (or 'FBP') given the sense in which it directly mimics the FBP approach proposed in Loaiza-Maya et al. (2021). Section 4 then illustrates the performance of loss-based ABF and FBP in an extensive set of simulation experiments under correctly-specified and misspecified SSMs. In particular, we assess the impact on predictive accuracy of different degrees of model misspecification and different forecast horizons. All results are compared with the accuracy of exact likelihood-based prediction, which is feasible in this artificial setting. The key result is that, in the case of marked model misspecification, both loss-based ABF and FBP produce 'coherent' predictions. That is, using an approach with a particular focus often yields the best out-of-sample performance according to that same measure of predictive accuracy. Moreover, the predictions produced by both focused methods are, more often than not, superior to those of the (misspecified) exact predictive. The conclusions are mixed when comparing the relative performance of loss-based ABF and FBP; however, the former approach often achieves better accuracy than the latter. As the forecast horizon increases, both loss-based methods retain the ability to generate coherent predictions in most instances; however, overall, coherence is more evident the shorter the forecast horizon.

Section 5 then applies both predictive methods to a stochastic volatility model with an intractable α-stable transition. Coherence is still a feature of both focused

methods. However, in this case, loss-based ABF outperforms FBP overall, suggesting that there are indeed benefits of conducting the focusing within an ABC algorithm that is driven - in part - by a well-specified SSM. The paper concludes in Section 6.

2. Bayesian forecasting in correctly-specified SSMs

2.1. Exact Bayesian forecasting

We assume a stationary ergodic process $\{Y_t\}_{t\geq 0}$ taking values in a measure space $(\mathcal{Y}, \mathcal{F}_Y)$, with $\mathcal{F}_Y$ a Borel σ-field, specified according to an SSM that depends on an unobserved state process $\{X_t\}_{t\geq 0}$, taking values in a measure space $(\mathcal{X}, \mathcal{F}_X)$, with $\mathcal{F}_X$ a Borel σ-field. Conditional on $\{X_t\}$, the sequence $\{Y_t\}$ is independent. To simplify the exposition, we choose to illustrate our approach in the case where both X_t and Y_t are continuous scalar random variables, with values x_t, y_t, $t = 1, \ldots, T$ and initial state x_1. For $t > 0$, we assume the following measurement and transition densities:

$$p(y_t|x_t, \theta) \tag{1}$$

$$p(x_t|x_{t-1}, \theta), \tag{2}$$

with an initial state density,

$$p(x_1|\theta), \tag{3}$$

where θ is a $P-$dimensional vector of unknown parameters. Whilst extension to the multivariate case (for either X_t or Y_t), with a concurrent increase in the dimension of θ, would cause no additional conceptual problems, ABC is not inherently well-suited to settings where θ is high-dimensional (Martin et al., 2024); hence our focus on the low-dimensional case in this paper. The first-order Markovian assumption for X_t is innocuous, and any finite (and known) Markov order can be accommodated by an appropriate definition of X_t.

For $p(\theta)$ defining the prior density, the posterior for θ can be expressed as

$$p(\theta|y_{1:T}) = \int_{\mathcal{X}^T} p(\theta, x_{1:T}|y_{1:T})\, dx_1 \cdots dx_T, \tag{4}$$

where $p(\theta, x_{1:T}|y_{1:T}) \propto p(y_{1:T}|x_{1:T}, \theta)p(x_{1:T}|\theta)p(\theta)$, with $x_{1:T} = (x_1, \ldots, x_T)'$ and $y_{1:T} = (y_1, \ldots, y_T)'$. In certain special cases (*e.g.* when (1) to (3) defines a linear Gaussian SSM), a Gibbs sampling scheme can be used to produce draws from $p(\theta, x_{1:T}|y_{1:T})$ and, thus, from $p(\theta|y_{1:T})$ (Carter & Kohn, 1994; Frühwirth-Schnatter, 1994). Typically, however, the conditionals $p(\theta|x_{1:T}, y_{1:T})$ and $p(x_{1:T}|\theta, y_{1:T})$ will not have known closed forms, and either a Metropolis Hasting (MH)-within-Gibbs scheme (Jacquier et al., 1994; Kim et al., 1998; Stroud et al., 2003; Strickland et al., 2006), or a PMCMC algorithm (Andrieu et al., 2011; Flury & Shephard, 2011) is applied. (See Giordani et al., 2011 and Jacquier & Polson, 2011 for reviews.) Once draws of θ and $x_{1:T}$ have been produced from $p(\theta, x_{1:T}|y_{1:T})$, the predictive probability density function (pdf),

$$p(y_{T+1}|y_{1:T}) = \int_{\mathcal{X}} \int_{\mathcal{X}^T} \int_{\Theta} p(y_{T+1}|x_{T+1}, \theta)p(x_{T+1}|x_T, \theta) \times p(\theta, x_{1:T}|y_{1:T})d\theta dx_{1:T} dx_{T+1}, \tag{5}$$

can be estimated using kernel density estimation methods, using subsequent draws from the transition and measurement densities, $p(x_{T+1}|x_T, \theta)$ and $p(y_{T+1}|x_{T+1}, \theta)$ respectively, or by averaging the measurement densities over the draws of x_{T+1} and θ.

However, this approach is operational only if both components of the (complete) likelihood function, $p(y_{1:T}|x_{1:T}, \theta)p(x_{1:T}|\theta)$, are available in closed form (for an MH-within-Gibbs scheme) or an unbiased estimator of each is available via particle filtering (for a PMCMC scheme). If neither of these conditions hold, draws from the augmented posterior, $p(\theta, x_{1:T}|y_{1:T})$, are unavailable and an estimate of $p(y_{T+1}|y_{1:T})$ that is exact up to simulation error cannot be produced. This is where ABC, and the associated production of an *approximation* of $p(y_{T+1}|y_{1:T})$, come into play.

2.2. Approximate Bayesian forecasting (ABF)

The aim of ABC is to produce draws from an approximation to $p(\theta|y_{1:T})$ in the case where $p(y_{1:T}|x_{1:T}, \theta)$, $p(x_{1:T}|\theta)$ (and $p(\theta)$) can be simulated from, even if $p(y_{1:T}|x_{1:T}, \theta)$ or $p(x_{1:T}|\theta)$ is unavailable in closed form. The simplest (accept/reject) form of the algorithm (Tavaré et al., 1997; Pritchard et al., 1999), adapted for the SSM, proceeds as per Algorithm 1. In Algorithm 1, $\eta(\cdot)$ is a (vector) statistic, $d\{\cdot, \cdot\}$ is a distance criterion, N is the number of draws from $p(\theta)$, and the tolerance level ε is chosen to be small.

Algorithm 1 ABC accept/reject algorithm

1: Simulate θ^i, $i = 1, 2, ..., N$, from $p(\theta)$.
2: Simulate the artificial data, $y^s_{1:T}(\theta^i)$, as follows:
- Simulate the states $x^s_{1:T}(\theta^i)$, $i = 1, ..., N$, from $p(x^s_1|\theta^i)$ and $p(x^s_t|x^s_{t-1}, \theta^i)$, for $t = 2, ..., T$.
- Simulate $y^s_{1:T}(\theta^i)$, $i = 1, ..., N$, from $p\left(y^s_t|x^s_t(\theta^i)\right)$, for $t = 1, ..., T$.

3: Select θ^i such that:

$$d\{\eta\left(y^s_{1:T}(\theta^i)\right), \eta(y_{1:T})\} \leq \varepsilon, \tag{6}$$

Denoting by $x^s_{1:T}$ and $y^s_{1:T}$, simulated draws from $p(x_{1:T}|\theta^i)$ and $p(y_{1:T}|x_{1:T}, \theta^i)$ respectively, an accepted draw of θ from Algorithm 1 is a draw from the posterior in equation 7 below, where $\mathbb{I}_\varepsilon[d\{\eta\left(y^s_{1:T}\right), \eta\left(y_{1:T}\right)\} \leq \varepsilon]$ is one if $d\{\eta\left(y^s_{1:T}\right), \eta\left(y_{1:T}\right)\} \leq \varepsilon$ and zero otherwise. When $\eta(\cdot)$ is sufficient for θ and for $\varepsilon \to 0$, $p_\varepsilon(\theta|\eta(y_{1:T}))$ is equivalent to the exact posterior, $p(\theta|y_{1:T})$. The sorts of intractable problems to which ABC is applied preclude sufficiency, almost by default; hence, ABC is only ever intrinsically approximate, even in cases where the computing budget allows the tolerance to be very small.[1]

[1] As is usual in practice, we apply a modified version of Algorithm 1, which is known as 'reference-table acceptance-rejection ABC' (*e.g.* Cornuet et al., 2008)., whereby we replace the acceptance step in Algorithm 1 with a nearest-neighbour selection step (Biau et al., 2015). The accepted draws of θ in this version of the algorithm are

$$p_\varepsilon(\theta|\eta(y_{1:T})) = \frac{\int_{y^s_{1:T}}\int_{x^s_{1:T}} \mathbb{I}_\varepsilon[d\{\eta\left(y^s_{1:T}\right),\eta\left(y_{1:T}\right)\}\le\varepsilon]p(y^s_{1:T}|x^s_{1:T},\theta)p(x^s_{1:T}|\theta)p(\theta)dx^s_{1:T}dy^s_{1:T}}{\int_\Theta\int_{y^s_{1:T}}\int_{x^s_{1:T}} \mathbb{I}_\varepsilon[d\{\eta\left(y^s_{1:T}\right),\eta\left(y_{1:T}\right)\}\le\varepsilon]p(y^s_{1:T}|x^s_{1:T},\theta)p(x^s_{1:T}|\theta)p(\theta)dx^s_{1:T}dy^s_{1:T}d\theta}. \tag{7}$$

Exploiting the Markov property of the state process in (2), the associated approximate predictive can be expressed as

$$\begin{aligned} &g(y_{T+1}|y_{1:T}) \\ &= \int_{\mathcal{X}}\int_{\mathcal{X}}\int_{\Theta} p(y_{T+1}|x_{T+1},\theta)p(x_{T+1}|x_T,\theta) \\ &\times\ p(x_T|\theta,y_{1:T})p_\varepsilon(\theta|\eta(y_{1:T}))d\theta dx_T dx_{T+1}. \end{aligned} \tag{8}$$

Given draws of θ from $p_\varepsilon(\theta|\eta(y_{1:T}))$ via Algorithm 1, all that is needed is a forward-filtering algorithm in order to draw from $p(x_T|\theta,y_{1:T})$ and to produce a simulation-based estimate of $g(y_{T+1}|y_{1:T})$, either as a sample mean of the measurement densities defined by the draws of θ (and x_{T+1}), or by applying kernel density techniques using subsequent draws from the transition and measurement densities, $p(x_{T+1}|x_T,\theta)$ and $p(y_{T+1}|x_{T+1},\theta)$. The key thing here is that posterior draws from the conditional posterior for the full vector of states is not required when it comes to implementing ABF. Also note that draws of x_T produced from Algorithm 1, which would implicitly be conditioned on $\eta(y_{1:T})$, not $y_{1:T}$, are not used. The particle filter enables draws of x_T to be conditioned on the complete data set, and this is critical for the resultant accuracy of $g(y_{T+1}|y_{1:T})$ as an approximation of $p(y_{T+1}|y_{1:T})$. (See Frazier et al., 2019, for further discussion.)

Given the obvious importance of the choice of summary statistics in determining how close $p_\varepsilon(\theta|\eta(y_{1:T}))$ can *ever* be to $p(\theta|y_{1:T})$, attention has been given to maximizing the information content of the summaries in some way (Joyce & Marjoram, 2008; Wegmann et al., 2009; Blum, 2010; Fearnhead & Prangle, 2012), including via the maximization of the likelihood function of an auxiliary model (Drovandi et al., 2011; Drovandi et al., 2015). It is the auxiliary-model approach to ABC that we pursue in this paper, given its superior performance in the SSM setting (Martin et al., 2019), plus the ease with which this method allows us to focus the ABC draws in the manner required in a loss-based setting. We will illustrate this latter point in detail below, once we have set the scene for the implementation of loss-based prediction *per se* in the case where the assumed SSM in (1) to (3) is a misspecified representation of the true DGP.

3. Bayesian forecasting in misspecified SSMs

3.1. Loss-based Bayesian prediction

The predictive distribution in (5), when accessed via an exact simulation method, is the gold standard in Bayesian prediction, and the benchmark against which any approximation – such as that in (8) – would be judged. However, its reliability as a benchmark depends critically on the assumption that the predictive model in (1) to (3) is correctly specified. The accuracy of the approximate predictive in (8) is also predicated on the assumption that the generating model used both in the ABC algorithm, and in the subsequent particle filtering algorithm, tallies with the true DGP (Martin et al., 2019). Once that assumption is violated, 'all bets are off', and the usefulness – or otherwise – of both (5) and (8) is solely a function of *how* misspecified the assumed model is.

In this case, it makes sense to take a different approach to prediction, namely to seek the form of predictive accuracy that matters for the problem at hand, rather than seeking correct model specification *per se*. More specifically, a more sensible predictive paradigm involves replacing the logarithmic scoring rule that explicitly underpins (5), and that is implicit in the generative model used in (8), by the *particular* scoring rule (and associated form of predictive accuracy) that matters for the particular forecasting problem being tackled. Such is the thinking that underpins the predictive methodology in Lacoste-Julien et al. (2011), Loaiza-Maya et al. (2021) and Frazier et al. (2024), and which we also adopt here.

We begin by assuming a class of plausible predictive SSMs for Y_{T+1}, conditioned on information $\mathcal{F}_T$, and indexed by the global parameter vector θ: $\mathcal{P}^{(T)} := \{P^{(T)}_\theta, \theta\in\Theta\}$. This class may comprise a single parametric SSM, in which θ retains its usual interpretation as the vector of parameters that underpins both the measurement and state distributions, or may comprise some combination of plausible models, in which case θ would consist of both the model-specific parameters and the combination parameters. To keep the scope of the paper manageable, we consider only the case of a class of single parametric models, and we continue to denote the measurement and transition densities of the assumed SSM using the notation in (1) to (3). The key thing is that we now no longer assume that the *true* predictive distribution is an element of $\mathcal{P}^{(T)}$.

Given the predictive class $\mathcal{P}^{(T)}$, the accuracy of $P^{(T)}_\theta \in \mathcal{P}^{(T)}$ can be measured using a positively-oriented proper scoring rule $s: \mathcal{P}^{(T)}\times\mathcal{Y}\mapsto\mathbb{R}$, where the expected scoring rule under the true distribution P_0 is defined as

$$\mathbb{S}(\cdot,P_0) := \int_{y\in\mathcal{Y}} s(\cdot,y)dP_0(y). \tag{9}$$

(See Gneiting & Raftery, 2007, for details of proper scoring rules). Since $\mathbb{S}(\cdot,P_0)$ cannot be attained in practice, a sample estimate based on $y_{1:T}$ is used to define the sample criterion for a given $\theta\in\Theta$, as

$$S_T(\theta) := T^{-1}\sum_{t=1}^{T} s(P^{(t)}_\theta, y_{t+1}). \tag{10}$$

associated with an empirical quantile over the simulated distances $d\{\eta\left(y^s_{1:T}\right),\eta\left(y_{1:T}\right)\}$. That is, Step 3 in Algorithm 1 is replaced with the following step: Select all θ^i associated with the $q=\delta/N$ smallest distances $d\{\eta\left(y^s_{1:T}\right),\eta\left(y_{1:T}\right)\}$ for some δ. We refer the reader to Martin et al. (2024) for an outline of the many further adaptations of the simple accept/reject ABC algorithm that have been proposed and for extensive referencing of the ABC literature.

Adopting the *generalized* updating rule proposed by Bissiri et al. (2016), amongst others, and first used by Loaiza-Maya et al. (2021) and Frazier et al. (2024) in a *forecasting* setting, we define

$$p_L(\theta|y_{1:T}) = \frac{\exp\left[wS_T(\theta)\right]\pi(\theta)}{\int_\Theta \exp\left[wS_T(\theta)\right]\pi(\theta)d\theta}, \tag{11}$$

where w is a scale factor, which needs to be set using certain criteria; see Loaiza-Maya et al. (2021) for details and relevant earlier references cited therein. In the spirit of the early work of Zhang (2006) and Jiang and Tanner (2008) this posterior may be referred to a *Gibbs* posterior. Alternatively, it may be termed a 'focused' posterior following Loaiza-Maya et al. (2021). Given that a negatively-oriented scoring rule can be viewed as a measure of loss in a predictive setting, we default to the term 'loss-based' in the main, and use the subscript L in (11) to signal this nomenclature.

By design, (11) places higher weight on elements of Θ that lead to predictive models, $P_\theta^{(T)}$, with higher predictive accuracy in the scoring rule $s(\cdot,\cdot)$ (or, equivalently, to models with lower predictive loss). As a result, the predictive defined by averaging with respect to $p_L(\theta|y_{1:T})$ rather than $p(\theta|y_{1:T})$ in (4):

$$\begin{aligned} &p_L(y_{T+1}|y_{1:T}) \\ &= \int_{\mathcal{X}}\int_{\mathcal{X}}\int_{\Theta} p(y_{T+1}|x_{T+1},\theta)p(x_{T+1}|x_T,\theta) \\ &\times\ p(x_T|\theta,y_{1:T})p_L(\theta|y_{1:T})d\theta\,dx_T\,dx_{T+1}, \end{aligned} \tag{12}$$

may well outperform, in the chosen rule $s(\cdot,\cdot)$, the predictive in (5) constructed using the exact – but misspecified – posterior in (4). Given its explicit dependence on the loss-based posterior, the predictive in (12) is referred to as the loss-based predictive (with the subscript L signalling this terminology), and with the additional adjective 'exact' also used whenever $p_L(y_{T+1}|y_{1:T})$ is accessed by an exact sampling scheme.

Superior predictive performance of an exact loss-based predictive has been established – both theoretically and numerically – in Loaiza-Maya et al. (2021), and similar validity is demonstrated in Frazier et al. (2024) for the predictive based on a variational Bayes approximation to $p_L(\theta|y_{1:T})$. However, both papers assume only *observation-driven* predictive classes. Our focus here is on the case where $\mathcal{P}^{(T)}$ is defined in terms of SSMs, and, as is evident from the inspection of (10), this causes an immediate problem. Specification of the sum of scores requires $P_\theta^{(t)}$ to be expressed as a function of θ alone, in order for the 'marginal' posterior, $p_L(\theta|y_{1:T})$, to be produced. This, in turn, requires the states to be integrated out. Whilst this could, in principle, be achieved in a preliminary step via numerical means, this is impractical from a computational point of view. Moreover, once the scenario of a truly intractable problem is entertained due to some component of the assumed model being unavailable analytically, $p_L(\theta|y_{1:T})$ is essentially out of reach. *Both* of these problems can be solved using ABC.

3.2. Loss-based ABF

In the spirit of Drovandi et al. (2011, 2015) and Martin et al. (2019), we implement an ABC approach by producing a vector of summary statistics, $\eta(\cdot)$, by *maximizing* a sample criterion function of the form of (10), but defined in terms of an auxiliary model chosen to be a reasonable representation of the assumed SSM in (1) to (3). A key assumption in our choice of auxiliary model is that it admits a closed-form predictive. This allows for the specification of a score-based criterion as:

$$S_T(\beta) := T^{-1}\sum_{t=1}^{T} s(P_\beta^{(t)}, y_{t+1}), \tag{13}$$

where $P_\beta^{(t)}$ denotes the conditional predictive associated with the auxiliary model, with parameter vector β. To keep the analysis manageable, we use four alternative forms of (positively-oriented) scoring rules:

$$s_{LS}\left(P_\beta^{(t)}, y_{t+1}\right) = \log p_\beta(y_{t+1}|y_{1:t}), \tag{14}$$

$$s_{CRPS}\left(P_\beta^{(t)}, y_{t+1}\right) = -\int_{-\infty}^{\infty}\left[P_\beta^{(t)} - I(y \geq y_{t+1})\right]^2 dy, \tag{15}$$

$$\begin{aligned} s_{CLS}\left(P_\beta^{(t)}, y_{t+1}\right) &= \log p_\beta(y_{t+1}|y_{1:t})I\left(y_{t+1} \in A\right) \\ &+ \left[\ln\int_{A^c} p_\beta(y|y_{1:t})dy\right] I\left(y_{t+1} \in A^c\right), \end{aligned} \tag{16}$$

$$\begin{aligned} s_{IS}\left(P_\beta^{(t)}, y_{t+1}\right) &= (u_{t+1} - l_{t+1}) + \frac{2}{\alpha}(l_{t+1} - y_{t+1}) \\ &\times\ I\left(y_{t+1} < l_{t+1}\right) + \frac{2}{\alpha}(y_{t+1} - u_{t+1})\, I\left(y_{t+1} > u_{t+1}\right), \end{aligned} \tag{17}$$

where $p_\beta(y_{t+1}|y_{1:t})$ represents the predictive density function associated with $P_\beta^{(t)}$, evaluated at the observed y_{t+1}. The log-score (LS) in (14) rewards a high value if the observed value, y_{t+1}, is in high density region of $p_\beta(.|y_{1:T})$, and maximization of (13) defined using the log-score yields the maximum likelihood estimator (MLE) of β. The continuously ranked probability score (CRPS) in (15) (see Gneiting & Raftery, 2007) is sensitive to distance and rewards the assignment of high predictive mass near to the realized y_{t+1}, rather than just at that value. The censored log-score (CLS) in (16), first proposed by Diks et al. (2011), rewards accuracy in a specific region, A (A^c denoting the complement), such as the tail of a predictive distribution for a financial return. In our numerical work, we define A as the lower and upper tail of the predictive distribution, as determined respectively by the 10% (or 20%) and 90% (or 80%) quantile of the empirical distribution of y_t, where we label these versions of CLS as respectively CLS10 (or CLS20) and CLS90 (or CLS80). Finally, the score in (17) is the interval score (IS) which is used in measuring the accuracy of a predictive interval. Here, we use it to measure the accuracy of a $100(1-\alpha)\%$ predictive interval where $\alpha = 0.05$. In (17), l_{t+1} and u_{t+1} denote the $100\left(\frac{\alpha}{2}\right)\%$ and $100\left(1-\frac{\alpha}{2}\right)\%$ predictive quantiles, respectively.

Define the jth scoring rule as $s_j(\cdot,\cdot)$, with $s_j, j \in \{\text{LS}, \text{CRPS}, \text{CLS10}, \text{CLS20}, \text{CLS80}, \text{CLS90}, \text{IS}\}$, $\hat{\beta}_j(y_{1:T})$ as the optimizer of (13) using s_j, and $y^s_{1:T}(\theta^i)$ as the vector of pseudo-data, $y^s_{1:T}$, produced using the ith draw θ^i from the prior $p(\theta)$. For any given rule s_j, the summary statistics $\eta\left(y^s_{1:T}\right)$ and $\eta\left(y_{1:T}\right)$ are defined as the first-derivative of (13) with respect to β computed, respectively, using the simulated and observed data, and with both evaluated at $\hat{\beta}_j(y_{1:T})$. That is:

$$\eta_j(y^s_{1:T}) = \bar{S}_j\left\{y^s_{1:T}(\theta^i); \hat{\beta}_j(y_{1:T})\right\}, \tag{18}$$

where:

$$\bar{S}_j\left\{y^s_{1:T}(\theta^i); \hat{\beta}_j(y_{1:T})\right\} = T^{-1}\sum_{t=1}^{T}\frac{\partial}{\partial\beta}\left[s_j\left(P^{(t)}_{\beta}, y^s_{t+1}(\theta^i)\right)\right]\Big|_{\beta=\hat{\beta}_j(y_{1:T})}$$

and:

$$\eta_j(y_{1:T}) = \bar{S}_j\left\{y_{1:T}; \hat{\beta}_j(y_{1:T})\right\} = 0, \tag{19}$$

since

$$\bar{S}_j\left\{y_{1:T}; \hat{\beta}_j(y_{1:T})\right\} = T^{-1}\sum_{t=1}^{T}\frac{\partial}{\partial\beta}\left[s_j(P^{(t)}_{\beta}, y_{t+1})\right]\Big|_{\beta=\hat{\beta}_j(y_{1:T})} = 0.$$

We take as the distance used in Algorithm 1 the Mahalanobis distance:

$$d\{\eta\left(y^s_{1:T}\right), \eta\left(y_{1:T}\right)\} = \sqrt{\left[\bar{S}_j\left\{y^s_{1:T}(\theta^i); \hat{\beta}_j(y_{1:T})\right\}\right]' \hat{\Sigma}\left[\bar{S}_j\left\{y^s_{1:T}(\theta^i); \hat{\beta}_j(y_{1:T})\right\}\right]}, \tag{20}$$

where $\hat{\Sigma}$ denotes the inverse of the (estimated) covariance matrix of $\eta_j(y^s_{1:T})$ across draws.

The accepted draws of θ produced by the (modified) Algorithm 1 (see Footnote 1) are then drawn from what we refer to as the loss-based ABC posterior, $p_{L,\varepsilon}(\theta|\eta(y_{1:T}))$ and, in turn, the ABC loss-based predictive can be defined as:

$$g_L(y_{T+1}|y_{1:T}) = \int_{\mathcal{X}}\int_{\mathcal{X}}\int_{\Theta} p(y_{T+1}|x_{T+1},\theta,y_{1:T})p(x_{T+1}|x_T,\theta)p(x_T|\theta,y_{1:T}) \times p_{L,\varepsilon}(\theta|\eta(y_{1:T}))d\theta dx_T dx_{T+1}. \tag{21}$$

It is the production and use of $g_L(y_{T+1}|y_{1:T})$ that we refer to hereafter as loss-based ABC prediction (or forecasting) or, for short, loss-based ABF.

3.3. Focused Bayesian prediction using the auxiliary model directly

The use of an auxiliary model with a closed-form predictive to drive the loss-based ABC prediction exercise does prompt one to explore an obvious alternative: namely, using the auxiliary model predictive *directly* in a generalized Bayesian update. Whilst such an approach avoids the use of a state space specification and, hence, cannot be used to conduct inference about the parameters of that process, it may well be that for the purpose of forecasting future observations generated from a state space model, use of the simpler model in a focused Bayesian update may be adequate. This is certainly the message gleaned from the numerical work undertaken in Loaiza-Maya et al. (2021), and this motivates our exploration of this alternative to loss-based ABF.

Use of the auxiliary model (with unknown parameter vector β) as the predictive model underpinning a focused update involves defining:

$$p_w(\beta|y_{1:T}) = \frac{\exp\left[wS_T(\beta)\right]p(\beta)}{\int_{\mathcal{B}}\exp\left[wS_T(\beta)\right]p(\beta)d\beta}, \tag{22}$$

where $S_T(\beta)$ is as defined in (13), and w is to be selected as described in Section 4. The one-step-ahead predictive is then constructed as:

$$p_{FBP}(y_{T+1}|y_{1:T}) = \int p_\beta(y_{t+1}|y_{1:t})p_w(\beta|y_{1:T})d\beta. \tag{23}$$

Since this principle of using a simple model to produce focused predictions in a misspecified setting directly mimics the 'focused Bayesian prediction' approach adopted in Loaiza-Maya et al. (2021), we use the abbreviation FBP hereafter to refer to the production and use of $p_{FBP}(y_{T+1}|y_{1:T})$.

4. Simulation exercise: Performance of loss-based ABF

In this section, we use an extensive set of simulation experiments to assess the accuracy of loss-based ABF under both correctly-specified and misspecified scenarios. We adopt as the assumed predictive model, an SSM that captures the main stylized features of a financial return, y_t, and which is specified in such a way that the exact (*i.e.* MCMC-based) predictive is available as a comparator. That is, while one of the prime motivations for using an ABC method is the presence of an intractable model – which, in turn, typically precludes the production of an exact predictive – in this simulation setting, we also wish to document the relationship between the approximate and exact predictive methods, particularly in the misspecified case. For this, we need to work with a tractable model.

In the correctly-specified case – *i.e.* when the true DGP matches this assumed model – the exact predictive is the gold standard, to which any alternative would be compared, including any approximate predictive. In the misspecified case, the exact predictive no longer assumes this status and the loss-based ABC predictives should – as they are designed to do – produce the best out-of-sample performance according to the score criterion on which they are based, and be superior to the misspecified exact predictive in anything other than log-score. Comparison is also made – under both scenarios – with FBP. Since the latter is based on a generalized update of the parameters of an auxiliary model that is *always* a simplified version of the assumed SSM for the return, the FBP results always reflect the impact of misspecification. As part of the exercise, we vary the degree of misspecification, documenting the effect of this on the performance of both loss-based ABF and FBP.

All results tabulated in this section are for one-step-ahead predictions. However, certain exercises were repeated for five- and ten-step-ahead predictive horizons. Those results are discussed briefly at the end of the section, but the tabulated results are included in the Appendix due to space constraints.

4.1. Simulation design

Here, the predictive class, $P_\theta^{(t)}$, is defined by an SSM, namely a simple stochastic volatility (SV) model for a continuously-compounded financial return, y_t, where α_t is the log volatility at time t:

$$y_t = \mu + e^{\alpha_t/2}e_t \quad ; \quad e_t \sim N(0,1) \tag{24}$$

$$\alpha_t = \bar{h}_\alpha + \phi(\alpha_{t-1} - \bar{h}_\alpha) + w_t \quad ; \quad w_t \sim N(0, \sigma_\alpha^2) \tag{25}$$

$$\alpha_1 \sim N\left(\bar{h}_\alpha, \frac{\sigma_\alpha^2}{1-\phi^2}\right), \tag{26}$$

with $\theta = (\phi, \sigma_\alpha^2, \mu, \bar{h}_\alpha)'$, where $\bar{h}_\alpha$ is the marginal mean of the log volatility, μ is the marginal mean of the return, and ϕ is the persistence of volatility. We adopt two alternative specifications for the DGP: (1) A model that matches the SV model in (24)–(26) (*i.e.* the assumed predictive model); and (2) An SV model that better replicates the stylized features of financial returns data, as used by Loaiza-Maya et al. (2021):

$$h_t = \bar{h} + a(h_{t-1} - \bar{h}) + \sigma_h \eta_t \tag{27}$$

$$z_t = e^{h_t/2}\epsilon_t \tag{28}$$

$$y_t = D^{-1}(F_z(z_t)), \tag{29}$$

where $\eta_t \overset{iid}{\sim} N(0,1)$ and $\epsilon_t \overset{iid}{\sim} N(0,1)$ are independent processes, $\{z_t\}_{t=1}^T$ is a latent process with stochastic variance $exp(h_t)$ and F_z is the implied marginal distribution of z_t, which is evaluated using simulation. An inverse distribution function associated with a standardized skewed-normal distribution D with shape parameter γ, is then used to produce the 'observed' return y_t as in (29), so that the data reflect the usual skewness observed in an empirical returns distribution, in addition to time-varying and autocorrelated volatility. Clearly, DGP (1) defines a correct-specification setting while DGP (2) defines a misspecification setting. As noted above, one of the objectives of the simulation exercise is to vary the level of misspecification and document the associated impact on the approximate predictive methods (loss-based ABF and its comparator, FBP). To achieve this, we generate data from DGP (2) with varying values of skewness via different values assigned to the parameter γ.

To generate the summary statistics for obtaining $p_{L,\varepsilon}(\theta|\eta(y_{1:T}))$ we adopt two different auxiliary models. It is necessary to choose an auxiliary model with a closed-form predictive. We have chosen our auxiliary models to meet this requirement while still being reasonable representations of the assumed SSM. The auxiliary models are defined as follows: (i) a Gaussian autoregressive conditionally heteroscedastic model of order one (ARCH(1)):

$$y_t = \beta_0 + \sqrt{\alpha_t}e_t \quad ; \quad e_t \overset{iid}{\sim} N(0,1) \quad ; \quad \alpha_t = \beta_1 + \beta_2(y_{t-1} - \beta_0)^2, \tag{30}$$

and (ii) a Gaussian generalized (G)ARCH(1,1) model:

$$\begin{aligned} y_t &= \beta_0 + \sqrt{\alpha_t}e_t \quad ; \quad e_t \overset{iid}{\sim} N(0,1) \quad ; \\ \alpha_t &= \beta_1 + \beta_2\alpha_{t-1} + \beta_3(y_{t-1} - \beta_0)^2. \end{aligned} \tag{31}$$

We take the following steps in producing the loss-based ABC one-step-ahead predictives (with appropriate adjustments made for producing the longer-horizon forecasts):

1. Generate T observations of y_t from the true DGP. Here, for the correct-specification scenario, we generate $T = 20000$ observations from (24)–(26) using parameter values: $(\phi, \sigma_\alpha, \mu, \bar{h}_\alpha) = (0.95, 0.3, 0.0009, -1.3)$. For the misspecification setting, $T = 20000$ observations are generated from (27)–(29) using a range of values for γ while fixing the other parameters. The fixed parameter values are: $(a, \bar{h}, \sigma_h) = (0.9, -0.4581, 0.4173)$. We adopt five different values for γ, namely -5, -4, -2, -1 and 0, thereby producing DGPs with decreasing levels of skewness. Given the symmetry of the assumed SSM in (24)–(26), and, indeed of the auxiliary models in (30) and (31), this generates a sequence of misspecification scenarios ranging from marked to mild.[2]
2. Use observations $t = 1, \ldots, 10000$ to evaluate $p_{L,\varepsilon}(\theta|\eta(y_{1:T}))$ using ABC. In the implementation of the ABC algorithm, the quantile used to select the number of draws from the prior (N) is allowed to decline as the sample size T increases, in accordance with the theoretical findings in Frazier et al. (2018). We use $q_T = 50T^{-3/2}$, which was also used by Martin et al. (2019) in their illustrations. Therefore, with 250 draws retained for density estimation, 5000000 replications are used to produce the results.

   (a) Correct-specification setting.
   Prior specifications: $\phi \sim U(0.5, 0.99)$, $\sigma_\alpha \sim U(0.05, 0.4)$, $\mu \sim N(0, 0.5)$, $\bar{h}_\alpha \sim N(-1, 1)$ are employed.

   (b) Misspecification setting.
   Prior specifications: $\phi \sim U(0.5, 0.99)$, $\sigma_\alpha \sim U(0.05, 0.4)$, $\mu \sim N(0, 1)$, $\bar{h}_\alpha \sim N(-3, 2)$ are employed.

   In both the correctly-specified and misspecified settings, the summary statistics and the distance criteria are specified as in (18)–(20), with all seven distinct scoring rules adopted, $s_j, j \in$ {LS, CRPS, CLS10, CLS20, CLS80, CLS90,

[2] We note here that the specification of $\gamma = 0$ still yields a DGP which differs from the assumed SSM and therefore characterizes a case of misspecification, albeit mild.

IS}. All scoring rules in (14)–(17) and the first derivatives of the score criteria have closed-form solutions for the two auxiliary models adopted here.

3. After obtaining $p_{L,\varepsilon}(\theta|\eta(y_{1:T}))$ using the first 10000 observations, we hold this posterior fixed, and use draws from it to construct the one-step-ahead predictives for the remaining 10000 time points as given in (21), using the bootstrap particle filter (Gordon et al., 1993), with 5000 particles drawn at each step. The average scores, based on the 'observed value', y_{t+1}, are then computed using $s_j, j \in$ {LS, CRPS, CLS10, CLS20, CLS80, CLS90, IS}. We use the notation "ABC-LS" to denote loss-based ABF in the case where the log-score is used in the criterion in (13). Corresponding abbreviations are used when (13) is constructed using other scoring rules.

When implementing the simulation design for FBP, we adopt as the predictive class, $P_\beta^{(t)}$, the same auxiliary model which is used for the comparable loss-based ABC predictive, and produce results under the six DGP scenarios described above. As highlighted earlier, *all* scenarios are settings of misspecification in this case, given the mismatch between the assumed predictive model underpinning FBP and all DGPs.

The following steps are undertaken in producing the FBP predictives:

1. Use observations $t = 1, \ldots, 10000$ to construct the FBP posterior in (22) using seven distinct scoring rules $s_j, j \in$ {LS, CRPS, CLS10, CLS20, CLS80, CLS90, IS}. For each predictive class and each score update, we produce (after thinning) $M = 4000$ posterior draws of $\theta, \theta^j, j = 1, 2, \ldots, M$ using a random walk MH (MCMC) algorithm.

   (a) Prior specifications for the Gaussian ARCH(1) model:
   Parameter vector: $\beta = (\beta_0, \beta_1, \beta_2)'$
   Prior: $\pi(\beta) \propto \frac{1}{\beta_1} \times I[\beta_1 > 0, \beta_2 \in [0, 1)]$

   (b) Prior specifications for the Gaussian GARCH(1,1) model:
   Parameter vector: $\beta = (\beta_0, \beta_1, \beta_2, \beta_3)'$
   Prior: $\pi(\beta) \propto \frac{1}{\beta_1} \times I[\beta_1 > 0, \beta_2 \in [0, 1), \beta_3 \in [0, 1)] \times I(\beta_2 + \beta_3 < 1)$

   When either (14) or (16) is used to define (22) a scale of $w = 1$ is adopted, which is considered a natural choice in these cases (see Loaiza-Maya et al., 2021). In the case where $w = 1$ and $s(P_\beta^{(t)}, y_{t+1}) = \log p_\beta(y_{t+1}|y_{1:t})$, the update in (22) obviously defaults to the conventional likelihood-based update of the prior defined over β, and the resultant predictive equates to the standard Bayesian predictive for the relevant auxiliary model. However, if (15) is used to determine the posterior update in (22), the interpretation of $\exp[wS_T(\beta)]$ as a pdf is no longer applicable, and the value of w must be chosen based on some criterion for weighting $\exp[wS_T(\beta)]$ and $\pi(\beta)$. As in Loaiza-Maya et al. (2021), we choose to target a value for w that ensures a rate of posterior update – when using CRPS – that is similar to that of the update based on LS, by defining

$$w = \frac{E_{p(\beta|y_{1:T})}\left[\sum_{t=1}^{T} s_{LS}(P_\beta^t, y_{t+1})\right]}{E_{p(\beta|y_{1:T})}\left[\sum_{t=1}^{T} s_{CRPS}(P_\beta^t, y_{t+1})\right]}.$$

The subscript $p(\beta|y_{1:T})$ indicates that the expectation is taken with respect to the exact posterior distribution for β. In practice, an estimate of w is produced as

$$\hat{w} = \frac{\sum_{j=1}^{J}\left[\sum_{t=1}^{T} s_{LS}(P_{\beta^{(j)}}^t, y_{t+1})\right]}{\sum_{j=1}^{J}\left[\sum_{t=1}^{T} s_{CRPS}(P_{\beta^{(j)}}^t, y_{t+1})\right]}, \tag{32}$$

using J draws of β from $p(\beta|y_{1:T})$, $\beta^j, j = 1, 2, \ldots, J$.
The same methodology is adopted when choosing the value for w when IS is used in the posterior update, and w is estimated in that case as

$$\hat{w} = \frac{\sum_{j=1}^{J}\left[\sum_{t=1}^{T} s_{LS}(P_{\beta^{(j)}}^t, y_{t+1})\right]}{\sum_{j=1}^{J}\left[\sum_{t=1}^{T} s_{IS}(P_{\beta^{(j)}}^t, y_{t+1})\right]}. \tag{33}$$

2. After obtaining the FBP posterior using the first 10000 observations, we hold this posterior fixed and use draws from it to construct the one-step-ahead predictives for the remaining 10000 time points. The average scores, based on the 'observed value', y_{t+1}, are then computed using $s_j, j \in$ {LS, CRPS, CLS10, CLS20, CLS80, CLS90, IS}. We label the method based on the use of scoring rules in (14)–(17) in the FBP update by FBP-LS, FBP-CRPS, FBP-CLS10, FBP-CLS20, FBP-CLS80, FBP-CLS90 and FBP-IS respectively.

As noted above, we also produce the exact predictive as a comparator for both loss-based ABF and FBP. The exact marginal posteriors are produced via a hybrid Gibbs-MH MCMC algorithm, where we apply the sparse matrix sampling algorithm of Chan and Jeliazkov (2009) to sample the states, conditional on θ, and a standard Gibbs algorithm to sample from the conditional posterior of θ given the states. Conditional on the draws of θ and α_T, draws of α_{T+1} are produced directly from $p(\alpha_{T+1}|\alpha_T, \theta)$, and the exact predictive is constructed by averaging the measurement densities over the draws of α_{T+1} and θ.

For reasons of space, in the sections that follow we report results related to the Gaussian GARCH(1,1) auxiliary model only, relegating specific results produced using the Gaussian ARCH(1) auxiliary model in the Appendix. Note that the qualitative nature of the conclusions we draw are the same for both auxiliary models.

Table 1
Predictive accuracy of loss-based ABF (Panel A) and FBP (Panel B) for one-step-ahead forecasts under the correct specification of the SSM. The Gaussian GARCH(1,1) model is adopted as the auxiliary model. The true DGP is the SV model in (24)–(26). The rows in Panel A refer to the scoring rule used in generating the summary statistics in the ABC algorithm underlying the loss-based ABF results. The rows in Panel B refer to the scoring rule used in FBP, while the last row in each panel refers to the exact predictive results. The columns in each panel refer to the scoring rule used to compute the out-of-sample average scores. The figures in bold are the largest average scores according to a given out-of-sample measure. The second largest average score is in italics.

Correct model specification							
	Panel A: Loss-based ABF Average out-of-sample score						
	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
Scoring rule							
ABC-LS	*−0.8189*	*−0.2956*	−0.4842	*−0.4875*	*−0.2985*	**−0.3211**	−3.0575
ABC-CLS10	−0.8244	−0.2964	−0.4857	−0.4907	−0.3016	*−0.3216*	−3.0946
ABC-CLS20	−0.8193	*−0.2956*	*−0.4841*	−0.4880	−0.2990	**−0.3211**	−3.0595
ABC-CLS80	−0.8192	−0.2957	**−0.4840**	−0.4879	−0.2988	**−0.3211**	−3.0595
ABC-CLS90	−0.8237	−0.2986	−0.4875	−0.4885	−0.2993	*−0.3216*	−3.0932
ABC-CRPS	**−0.8188**	**−0.2955**	*−0.4841*	**−0.4874**	**−0.2983**	**−0.3211**	**−3.0551**
ABC-IS	**−0.8188**	**−0.2955**	**−0.4840**	−0.4876	−0.2986	**−0.3211**	*−3.0559*
Exact	−0.8191	−0.2957	−0.4842	−0.4876	−0.2986	−0.3211	−3.0594
	Panel B: FBP Average out-of-sample score						
	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
Scoring rule							
FBP-LS	*−0.8557*	−0.3119	−0.5020	−0.5009	−0.3111	*−0.3238*	*−3.1525*
FBP-CLS10	−0.8989	**−0.3046**	**−0.4934**	−0.5425	−0.3380	−0.3363	−3.2309
FBP-CLS20	−0.8838	*−0.3054*	*−0.4941*	−0.5287	−0.3274	−0.3322	−3.1963
FBP-CLS80	−0.8862	−0.3273	−0.5315	*−0.4959*	*−0.3061*	−0.3328	−3.2051
FBP-CLS90	−0.9154	−0.3470	−0.5579	**−0.4958**	**−0.3051**	−0.3408	−3.2699
FBP-CRPS	−0.8616	−0.3165	−0.5057	−0.5055	−0.3162	**−0.3228**	−3.1930
FBP-IS	**−0.8551**	−0.3099	−0.5011	−0.4997	−0.3094	−0.3242	**−3.1434**
Exact	−0.8191	−0.2957	−0.4842	−0.4876	−0.2986	−0.3211	−3.0594

4.2. Simulation results: Correct specification

In Panel A of Table 1 we record the predictive accuracy of loss-based ABF for one-step-ahead forecasts under the correct specification of the SSM. The rows in Panel A refer to the scoring rule used to generate the summary statistics in the ABC algorithm, and the columns record the average out-of-sample loss according to the score nominated in the column heading.

Given the use of proper scoring rules to define the relevant criterion function in (13), we would anticipate that, under correct specification, the optimization process would produce essentially equivalent values for the summary statistics used in the ABC algorithm and, hence, (essentially) equivalent draws of θ. This, in turn, would lead to ABC-based posteriors that concentrate onto the same parameter value for a large enough sample, and predictive results that are indistinguishable across all scoring rules as a consequence. This is what is observed in Panel A, with the slightly superior forecasting performance of ABC-CRPS in several instances resulting from finite sample fluctuations that abate as the number of in-sample observations increases. Moreover, and as tallies with the asymptotic and numerical analysis in Frazier et al. (2019), the performance of the log-score-based ABC algorithm – virtually identical to all other versions of the algorithm, as per the above argument – is also virtually equivalent to that of the exact Bayesian method for all out-of-sample scores.

In summary, when the SSM used in the ABC algorithm is correctly specified, and contingent on the sample size being sufficiently large, use of the likelihood function of an auxiliary model to produce the summary statistics is all that is required to produce predictive results that are equivalent in accuracy to those produced via the exact predictive, and according to any measure of out-of-sample accuracy. Focusing on a particular form of predictive accuracy, via the specification of a different criterion function within ABC, reaps no additional benefit.

In contrast, if one adopts the approach used in Loaiza-Maya et al. (2021) and focuses the predictives *directly* via the use of the Gaussian GARCH(1,1) model in a generalized Bayesian update, the different updates do lead to different predictives, as is evident from the results recorded in Panel B of Table 1. In particular, the expected appearance of the largest average scores in diagonal positions (highlighted in the bold font) is in evidence overall; that is, using FBP with a particular focus yields the best out-of-sample performance according to that same measure of predictive accuracy or, using the nomenclature of Martin et al. (2022), yields 'coherent' predictions. This is simply the consequence of the fact that – by construction

Table 2
Predictive accuracy of loss-based ABF (Panel A) and FBP (Panel B) for one-step-ahead forecasts under misspecification of the SSM. The Gaussian GARCH(1,1) model is adopted as the auxiliary model. The true DGP is the SV model in (27)–(29) with $\gamma = -5$, whilst the assumed DGP underlying loss-based ABF is the SV model in (24)–(26). The rows in Panel A refer to the scoring rule used in generating the summary statistics in the ABC algorithm underlying the loss-based ABF results. The rows in Panel B refer to the scoring rule used in FBP, while the last row in each panel refers to the exact (but misspecified) predictive results. The columns in each panel refer to the scoring rule used to compute the out-of-sample average scores. The figures in bold are the largest average scores according to a given out-of-sample measure. The second largest average score is in italics.

Model misspecification							
	Panel A: Loss-based ABF Average out-of-sample score						
	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
Scoring rule							
ABC-LS	*−1.3427*	−0.3586	**−0.6173**	−0.4900	−0.2975	*−0.5331*	−4.6333
ABC-CLS10	−1.4117	*−0.3572*	−0.6327	−0.5122	−0.3037	−0.5616	*−4.5813*
ABC-CLS20	−1.3737	**−0.3553**	*−0.6202*	−0.5062	−0.3084	−0.5427	−4.7791
ABC-CLS80	−2.0917	−0.8118	−1.2925	**−0.4675**	*−0.2822*	−0.6082	−10.5000
ABC-CLS90	−2.4259	−0.8961	−1.4896	*−0.4715*	**−0.2777**	−0.6509	−12.7820
ABC-CRPS	**−1.3371**	−0.3629	−0.6214	−0.4881	−0.2998	**−0.5309**	−4.7405
ABC-IS	−1.4882	−0.3657	−0.6648	−0.5333	−0.3057	−0.6025	**−4.2895**
Exact	−1.3343	−0.3618	−0.6199	−0.4882	−0.3003	−0.5304	−4.7357

	Panel B: FBP Average out-of-sample score						
	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
Scoring rule							
FBP-LS	**−1.3471**	−0.3694	−0.6312	−0.4840	−0.2954	*−0.5340*	*−4.6822*
FBP-CLS10	−1.3669	**−0.3593**	*−0.6243*	−0.5094	−0.3212	−0.5380	−4.8663
FBP-CLS20	−1.4076	*−0.3601*	**−0.6201**	−0.5606	−0.3749	−0.5464	−5.2801
FBP-CLS80	−2.0718	−0.9227	−1.3493	**−0.4491**	*−0.2657*	−0.5888	−8.9718
FBP-CLS90	−2.5938	−1.1223	−1.7430	*−0.4579*	**−0.2644**	−0.6357	−11.5340
FBP-CRPS	*−1.3485*	−0.3786	−0.6378	−0.4839	−0.2973	**−0.5319**	−4.7954
FBP-IS	−1.4521	−0.3734	−0.6765	−0.5106	−0.2943	−0.5979	**−4.3179**
Exact	−1.3343	−0.3618	−0.6199	−0.4882	−0.3003	−0.5304	−4.7357

– the auxiliary model is misspecified, and it is under such misspecification that focusing reaps benefits.

However, and as is critical to note in the current context, one is still much better off using a correctly-specified SSM within an ABC algorithm, despite the approximate nature of the algorithm, than resorting to an FBP approach (implemented via an exact MCMC algorithm) using a misspecified predictive model. That is, all of the average scores in Panel B for the FBP methods are notably lower than the corresponding scores for the loss-based ABF methods in Panel A and, in turn, lower than the corresponding results for the exact predictive. This is really the key conclusion to draw from the numerical results in Table 1.

4.3. Simulation results: Misspecification

As noted, one of our goals is to explore the impact on both loss-based ABF and FBP of varying levels of model misspecification by allowing for a range of different values of γ in the DGP. We record results corresponding to three values for γ in the main text, $\gamma = -5, -4, 0$, with the results for $\gamma = -2$ and $\gamma = -1$ available in the Appendix.

Panel A in Table 2 documents the predictive accuracy of loss-based ABF for one-step-ahead forecasts when the DGP corresponds to the SV model in (27)–(29) with $\gamma = -5$, which – by having the largest degree of negative skewness – is considered the most misspecified case. As anticipated, we now observe differences in predictive accuracy within the columns, which highlights the fact that the different scoring rules used in the ABC mechanism to generate the summary statistics are producing different predictives. Further, we can observe that, with minor deviations, the expected appearance of bold figures on the main diagonals is in evidence; that is, using ABC with a particular predictive focus in the auxiliary criterion function yields the best out-of-sample performance according to that same measure of predictive accuracy, overall. Moreover, in several cases, despite the approximate nature of the ABC predictives, the focusing produces more accurate results out-of-sample – according to the relevant scoring rule – than the exact (and now misspecified) predictive. The most obvious exception to this statement is the italicized result for ABC-LS in the first column of Panel A, which is no better than the exact predictive result. This is not surprising since both predictives are driven by the log score in this case, and superiority of the approximate predictive would not be anticipated.

Table 3
Predictive accuracy of loss-based ABF (Panel A) and FBP (Panel B) for one-step-ahead forecasts under misspecification of the SSM. The Gaussian GARCH(1,1) model is adopted as the auxiliary model. The true DGP is the SV model in (27)–(29) with $\gamma = -4$, whilst the assumed DGP underlying loss-based ABF is the SV model in (24)–(26). The rows in Panel A refer to the scoring rule used in generating the summary statistics in the ABC algorithm underlying the loss-based ABF results. The rows in Panel B refer to the scoring rule used in FBP, while the last row in each panel refers to the exact (but misspecified) predictive results. The columns in each panel refer to the scoring rule used to compute the out-of-sample average scores. The figures in bold are the largest average scores according to a given out-of-sample measure. The second largest average score is in italics.

Model misspecification							
	Panel A: Loss-based ABF Average out-of-sample score						
	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
Scoring rule							
ABC-LS	*−1.3423*	−0.3559	**−0.6130**	−0.4989	−0.3017	*−0.5331*	*−4.5893*
ABC-CLS10	−1.4145	*−0.3557*	−0.6307	−0.5207	−0.3062	−0.5651	−4.4636
ABC-CLS20	−1.3678	**−0.3532**	*−0.6154*	−0.5122	−0.3109	−0.5409	−4.7258
ABC-CLS80	−1.9576	−0.7594	−1.1787	**−0.4880**	*−0.2963*	−0.5958	−9.4963
ABC-CLS90	−2.1688	−0.7993	−1.3105	*−0.4910*	**−0.2941**	−0.6284	−11.2290
ABC-CRPS	**−1.3368**	−0.3593	−0.6161	−0.4976	−0.3042	**−0.5310**	−4.6903
ABC-IS	−1.4647	−0.3612	−0.6520	−0.5353	−0.3094	−0.5916	**−4.3072**
Exact	−1.3344	−0.3586	−0.6153	−0.4968	−0.3036	−0.5307	−4.6817
	Panel B: FBP Average out-of-sample score						
	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
Scoring rule							
FBP-LS	**−1.3456**	−0.3661	−0.6261	−0.4920	−0.2984	*−0.5340*	*−4.6454*
FBP-CLS10	−1.3629	**−0.3572**	*−0.6202*	−0.5135	−0.3201	−0.5374	−4.8076
FBP-CLS20	−1.3977	*−0.3580*	**−0.6163**	−0.5576	−0.3656	−0.5442	−5.1827
FBP-CLS80	−1.8378	−0.7711	−1.1281	**−0.4660**	*−0.2761*	−0.5743	−7.8388
FBP-CLS90	−2.1744	−0.9385	−1.4055	*−0.4715*	**−0.2752**	−0.6089	−9.6594
FBP-CRPS	*−1.3467*	−0.3738	−0.6314	−0.4922	−0.3001	**−0.5321**	−4.7432
FBP-IS	−1.4332	−0.3688	−0.6641	−0.5139	−0.2981	−0.5876	**−4.3422**
Exact	−1.3344	−0.3586	−0.6153	−0.4968	−0.3036	−0.5307	−4.6817

In Panel B, once again we see the coherence of the FBP results and more uniformly than was observed in Panel B of Table 1. Given that the Gaussian GARCH(1,1) model is arguably even more misspecified in this case, with the true DGP including skewness, this is simply evidence of the fact that (subject to caveats) focusing reaps more benefits the more extreme is the misspecification (see Loaiza-Maya et al., 2021, and Martin et al., 2022). As with the loss-based ABF results, in several cases focusing on the Gaussian GARCH(1,1) model yields superior out-of-sample performance than using the exact but misspecified SSM.

Most importantly, when one compares the results in Panels A and B, the conclusions are mixed. We observe that FBP is doing better than loss-based ABF in terms of accuracy in the upper tails, as measured by CLS80 and CLS90. These results relating to FBP are consistent with the findings by Loaiza-Maya et al. (2021), produced under the same simulation scenario, where FBP is found to reap particular benefits in the upper tail. In contrast, ABC-CRPS and ABC-IS outperform their FBP counterparts. We also observe that loss-based ABF performs slightly better than FBP when it is driven by LS and CLS10, but only by a small margin. To three decimal places, the ABC and FBP results based on the CLS20 criterion are equivalent, in terms of average out-of-sample CLS20.[3]

Tables 3 and 4 document the predictive accuracy of loss-based ABF and FBP under reduced misspecification. Specifically, the SV model defined in (27)–(29) is employed as the DGP, with $\gamma = -4$ (Table 3) and $\gamma = 0$ (Table 4) adopted. As is evident in Table 3, coherent predictions still feature for both methods despite the slightly less marked skewness in the DGP compared to the case in Table 2. In several cases, the best-performing focused method is still superior to the exact but misspecified predictive. Comparing the two panels in Table 3, loss-based ABF performs slightly better than FBP when driven by LS, CLS10, CLS20, CRPS and IS, while FBP still outperforms loss-based ABF in the upper tails, as measured

[3] In response to a request from a referee, we have documented (for this particular misspecification setting) the computing time for both focused methods, loss-based ABF and FBP, recording those times in Table 8 in the Appendix. The time taken to compute a single loss-based ABC predictive is longer than for an FBP predictive due to the time taken to produce the reference table (of posterior draws) for ABC and the need to run a particle filtering step to produce the ABC-based predictives. However, regardless of the method chosen, computing times are similar across the various scoring rules used in the focusing.

Table 4
Predictive accuracy of loss-based ABF (Panel A) and FBP (Panel B) for one-step-ahead forecasts under misspecification of the SSM. The Gaussian GARCH(1,1) model is adopted as the auxiliary model. The true DGP is the SV model in (27)–(29) with $\gamma = 0$, whilst the assumed DGP underlying loss-based ABF is the SV model in (24)–(26). The rows in Panel A refer to the scoring rule used in generating the summary statistics in the ABC algorithm underlying the loss-based ABF results. The rows in Panel B refer to the scoring rule used in FBP, while the last row in each panel refers to the exact (but misspecified) predictive results. The columns in each panel refer to the scoring rule used to compute the out-of-sample average scores. The figures in bold are the largest average scores according to a given out-of-sample measure. The second largest average score is in italics.

	Model misspecification						
	Panel A: Loss-based ABF Average out-of-sample score						
	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
Scoring rule							
ABC-LS	**−1.3441**	*−0.3254*	*−0.5571*	*−0.5587*	*−0.3301*	*−0.5342*	*−4.2999*
ABC-CLS10	−1.5235	−0.3270	−0.5697	−0.6740	−0.4022	−0.5882	−5.2434
ABC-CLS20	−1.4136	**−0.3236**	**−0.5567**	−0.6066	−0.3611	−0.5555	−4.5544
ABC-CLS80	−1.4178	−0.3617	−0.6129	**−0.5586**	**−0.3291**	−0.5541	−4.6618
ABC-CLS90	−1.5330	−0.4063	−0.6870	−0.5733	−0.3348	−0.5854	−5.4236
ABC-CRPS	*−1.3465*	−0.3279	−0.5595	−0.5605	−0.3322	**−0.5335**	−4.4644
ABC-IS	*−1.3465*	*−0.3254*	−0.5573	−0.5592	−0.3305	−0.5358	**−4.2497**
Exact	−1.3391	−0.3239	−0.5553	−0.5566	−0.3285	−0.5340	−4.3008
	Panel B: FBP Average out-of-sample score						
	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
Scoring rule							
FBP-LS	**−1.3376**	−0.3235	*−0.5549*	−0.5560	−0.3278	**−0.5344**	*−4.3566*
FBP-CLS10	−1.3566	**−0.3227**	−0.5561	−0.5703	−0.3395	−0.5412	−4.4292
FBP-CLS20	−1.3407	*−0.3230*	**−0.5547**	−0.5584	−0.3299	−0.5357	−4.3589
FBP-CLS80	−1.3412	−0.3274	−0.5589	*−0.5556*	*−0.3270*	−0.5353	−4.3608
FBP-CLS90	−1.3563	−0.3377	−0.5721	−0.5565	**−0.3263**	−0.5400	−4.4420
FBP-CRPS	−1.3391	−0.3246	−0.5557	−0.5571	−0.3288	**−0.5344**	−4.3931
FBP-IS	*−1.3387*	*−0.3230*	−0.5550	**−0.5555**	*−0.3270*	*−0.5351*	**−4.3156**
Exact	−1.3391	−0.3239	−0.5553	−0.5566	−0.3285	−0.5340	−4.3008

by CLS80 and CLS90. These same qualitative results also hold for the case of $\gamma = -2$ (documented in Table 9 in the Appendix). Once misspecification becomes milder, as characterized by $\gamma = 0$, the discrepancies in predictive accuracy across the values in any given column become much less marked, as do any improvements in focusing over exact prediction. Under this particular form of (mild) misspecification, loss-based ABF outperforms FBP only when driven by CRPS and IS and as measured by the corresponding average score. The results in Table 10, for $\gamma = -1$, (in the Appendix), are qualitatively similar to those in Table 4.

Overall then, in the case of marked model misspecification, loss-based ABF and FBP produce coherent predictions, and predictions that are better than those of the (misspecified) exact predictive. In terms of the dominance of one method over the other, conclusions are mixed under different misspecification scenarios. However, regardless of the level of misspecification, loss-based ABF consistently exhibits superior out-of-sample performance compared to FBP when driven by CRPS and IS, while FBP consistently performs better in the upper tails as measured by CLS80 and CLS90. Hence, whilst focusing within the ABC algorithm can reap benefits relative to a more crude form of direct update via the auxiliary model itself, this dominance may not be uniform, and is likely to depend on the extent to which the assumed SSM is misspecified. That said, there is little to lose from going the ABC route, and the motivation may be stronger once the SSM being entertained is truly intractable. This is explored as part of the empirical exercise to follow.

Finally, to gauge the impact on forecast accuracy of longer forecast horizons, we repeated the simulation exercise using the SV model defined in (27)–(29) with $\gamma = -5$ to produce five-step-ahead and ten-step-ahead forecasts. As noted earlier, the relevant predictive accuracy results are presented in Tables 11 and 12 in the Appendix. Interestingly, under these longer forecast horizons, both methods still demonstrate the ability to produce coherent predictions in most instances, although the coherence is less consistent than in the one-step-ahead case. Again, the conclusions are mixed in terms of the dominance of one method over the other.

5. Empirical illustration: SV model with an intractable transition

In this section, we conduct a comparative analysis of the accuracy of the two predictive methods, loss-based ABF and FBP, in a simple empirical setting where we

forecast the daily returns on the S&P500 index. Here, we adopt an SV model with α-stable transitions as the assumed predictive model, which is now a truly intractable SSM, for which exact Bayesian prediction – even if deemed to be desirable, given the likelihood of misspecification – is infeasible.

5.1. Models and computational details

There are several empirical studies in which the non-Gaussian features of financial returns are captured via the use of α-stable processes (Carr & Wu, 2003; Lombardi & Calzolari, 2009; Peters et al., 2012; Frazier et al., 2019; Martin et al., 2019). With the α-stable process not possessing a closed-form representation for the density function, the exact Bayesian predictive is not accessible. ABC-based prediction (and inference), on the other hand, *is* feasible, given that the α-stable process *can* be simulated via the algorithm proposed in Chambers, Mallows and Stuck (1976), and such has been the motivation for the use of ABC in Peters et al. (2012), Frazier et al. (2019) and Martin et al. (2019). In this spirit, we adopt here an α-stable process to drive the innovation to the (log) volatility of the financial return.

Specifically, we define the following model for the continuously-compounded return, y_t, on the S&P500 index:

$$y_t = e^{h_t/2} e_t \quad ; \quad e_t \overset{iid}{\sim} N(0,1) \tag{34}$$

$$h_t = \omega + \phi h_{t-1} + \sigma_h \eta_t \quad ; \quad \eta_t \overset{iid}{\sim} S(\alpha, \gamma, \mu, \sigma), \tag{35}$$

where h_t is the log volatility at time t, and $S(\alpha, \gamma, \mu, \sigma)$ denotes an α-stable Lévy process with location parameter μ, scale parameter σ, tail index $\alpha \in (1, 2)$, and skewness parameter γ. The values of α and γ control the degree of leptokurtosis and skewness (respectively) in the innovations to the log volatility process, and we choose to fix the degree of (negative) skewness in the model via the specification of $\gamma = -1$. We also set $\mu = 0$ and $\sigma = 1$.

We use daily close-to-close returns data (sourced from Global Financial Data) from 4 January 2010 to 31 December 2019, comprising 2516 observations. The most recent 500 observations are reserved for the one-step-ahead predictive assessments, with the initial set of 2016 observations used to produce the relevant posterior for the unknown parameters, $\theta = (\omega, \phi, \sigma_h, \alpha)'$, from which draws are taken: ABC draws in the case of loss-based ABF and MCMC draws in the case of FBP. That is, as in the simulation exercise, the posterior is produced only once and not updated over the out-of-sample period.

We implement the loss-based ABF approach in this empirical setting with the same summary statistics and distance criterion as specified in (18)–(20). In the context of financial returns, accurate prediction of extreme returns, whether very large or very small, is particularly important. Hence, in addition to LS, we opt for CLS, which rewards predictive accuracy in a tail, while excluding the use of CRPS and IS for this exercise. Specifically, we employ five distinct scoring rules denoted as $s_j, j \in \{\text{LS}, \text{CLS10}, \text{CLS20}, \text{CLS80}, \text{CLS90}\}$ to define the relevant sample criterion in (13). We use the Gaussian GARCH(1,1) model as defined in (31) as the auxiliary model, and employ uniform priors: $\omega \sim U(-1, 1)$, $\phi \sim U(0.5, 0.99)$, $\sigma_h \sim U(0, 0.3)$, $\alpha \sim U(1, 2)$. FBP is implemented via direct use of the same Gaussian GARCH(1,1) model as the assumed predictive model in (22), with the priors for the parameters of the GARCH model the same as specified in Section 4.1.

5.2. Empirical forecasting results

Table 5 documents the empirical predictive results for the two methods. Panel A records the average out-of-sample scores for the loss-based ABF method, while Panel B records the corresponding results for FBP.

The results show that, apart from minor deviations, the bold figures align as expected along the main diagonal of each panel. Thus, both predictive mechanisms can produce coherent predictions in this empirical setting. That is, focusing the predictive mechanism on the scoring rule that matters out-of-sample – either via the ABC machinery or via a direct update using the auxiliary model – does reap benefits.

However, when it comes to the predictive *superiority* of one approach over another, the findings are once again mixed, mirroring the results in the (misspecification) simulation exercise in Section 4.3. As a comparison of the corresponding bolded results in Panels A and B of Table 5 indicates, loss-based ABF dominates in terms of both log-score and upper-tail-based accuracy, whilst FBP outperforms loss-based ABF in terms of lower tail accuracy. Interestingly, and not surprisingly, the use of the well-specified SSM in the ABC algorithm leads to the predictive scores in any one particular column of Panel A being much less diverse than the numbers in the corresponding column in Panel B, and higher – sometimes much higher – in most cases. That is, the ABC approach, in exploiting a more flexible – and arguably less misspecified – model than does the FBP approach, produces higher out-of-sample scores, on average, as a consequence. In summary, whilst the best version of FBP may be better than the best version of loss-based ABF in the lower tail, *overall* the average scores in Panel A are higher than those in Panel B, highlighting the importance of factoring in a well-specified predictive model, and the benefits of using ABC to do so.

6. Discussion

We have developed a new approach for conducting Bayesian prediction in state space models (SSMs) that does not rely on correct model specification and which accommodates model intractability. Termed loss-based approximate Bayesian forecasting (loss-based ABF), a posterior is constructed using an approximate Bayesian computation (ABC) algorithm in which the (vector) summary statistic is produced by maximizing a criterion function – defined, in turn, using a closed-form auxiliary predictive – that rewards a user-specified measure of predictive accuracy. The resultant predictive – by construction – may yield more accuracy out-of-sample when accuracy is

Table 5
Predictive accuracy of loss-based ABF (Panel A) and FBP (Panel B) for one-step-ahead forecasts in an empirical setting. The Gaussian GARCH(1,1) model is adopted as the auxiliary model. The assumed DGP underlying loss-based ABF is the SV model in (34)–(35). The rows in Panel A refer to the scoring rule used in generating the summary statistics in the ABC algorithm. The rows in Panel B refer to the scoring rule used in updating FBP. The columns in each panel refer to the out-of-sample measure used to compute the average scores. The figures in bold are the largest average scores according to a given out-of-sample measure. The second largest average score is in italics.

Assumed predictive model: SV model with α-stable errors					
	Panel A: Loss-based ABF Average out-of-sample score				
	LS	CLS10	CLS20	CLS80	CLS90
Scoring rule					
ABC-LS	3.3967	0.0268	0.2923	0.5170	0.1523
ABC-CLS10	**3.4058**	**0.0391**	**0.3048**	0.5122	0.1500
ABC-CLS20	*3.4053*	*0.0383*	*0.3041*	0.5126	0.1502
ABC-CLS80	3.3772	0.0052	0.2713	**0.5192**	**0.1547**
ABC-CLS90	3.3849	0.0134	0.2790	*0.5191*	*0.1540*

	Panel B: FBP Average out-of-sample score				
	LS	CLS10	CLS20	CLS80	CLS90
Scoring rule					
FBP-LS	**3.3467**	−0.0262	0.2406	0.4801	0.1473
FBP-CLS10	3.1051	**0.0463**	**0.3096**	0.1903	−0.1119
FBP-CLS20	3.1788	*0.0417*	*0.3085*	0.2645	−0.0430
FBP-CLS80	*3.3293*	−0.0515	0.2155	**0.4860**	*0.1531*
FBP-CLS90	3.1941	−0.1204	0.1042	*0.4820*	**0.1543**

assessed using that particular measure. Two comparators are entertained: exact (but misspecified) Bayesian prediction (available in a simulation setting at least) and prediction based on a generalized Bayesian update using the auxiliary model alone. We refer to the latter as focused Bayesian prediction (FBP), given that it directly mimics the approach with the same name proposed in Loaiza-Maya et al. (2021).

The simulation results reveal that in the case of marked model misspecification, the new approach can yield both coherent predictions, and predictions that are more accurate than the exact (but misspecified) MCMC-based predictions and often (if not uniformly) more accurate than the FBP results. An interesting fact is that, under model misspecification, the proposed approach can still produce coherent predictions for longer horizons. However, coherence is more consistent when the forecast horizon is shorter. In an empirical setting, using an intractable SSM within the ABC algorithm, and for which no exact comparator is available, loss-based ABF is almost always more accurate than FBP – exploiting as it does, an empirically realistic and flexible SSM as part of the ABC algorithm. However, that said, the very best FBP result – in terms of accurate returns prediction in the lower tail – is superior to the corresponding loss-based ABF result.

The overall conclusion that we draw is that there is little to lose from adopting the ABC-based approach, and much to gain. As is consistent with previous conclusions about the robustness of predictive accuracy to any inaccuracy in an ABC posterior (Frazier et al., 2019), the approximate nature of the loss-based ABF predictives impinges little on predictive accuracy. This fact, allied with the ability of the algorithm to *both* focus on the type of predictive accuracy that matters and to accommodate intractable state space specifications, makes it an attractive proposition. That said, in some cases, the more straightforward and direct focusing via the FBP approach may still be entertained, in particular when the specification of a realistic SSM within an ABC algorithm is deemed to be challenging; but we would view this option as being the less desirable one, as a general rule.

As a final point, we note that the proposed ABC procedure can be viewed as a type of likelihood-free 'cutting feedback' approach, whereby the flow of information from the latent states to the static parameters, which drive the dynamics of the latent states, is suppressed entirely.[4] In this context, we learn about the static parameters via ABC and the specific summaries that drive predictive performance, without conducting inference on the latent states; inference on the states is then conducted conditional on the parameters, and the full set of data, via the bootstrap filter. Thus, inference about the parameters impacts the states, but not the converse, in contrast to the usual Bayesian treatment of SSMs. We conjecture that there may be a link between positive forecast performance in misspecified SSMs, and cutting the feedback between the states and the static parameters, as is occurring in the loss-based ABC method; however, we leave a detailed analysis of this conjecture for future research.

[4] Cutting feedback is a recently proposed method for producing robust Bayesian inferences under model misspecification; see Nott et al. (2023) for a recent review and discussion, and see Chakraborty et al. (2023) for an application of cutting feedback to forecasting in likelihood-free settings.

CRediT authorship contribution statement

Chaya Weerasinghe: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Rubén Loaiza-Maya:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Gael M. Martin:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **David T. Frazier:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

This appendix documents:

(i) The predictive accuracy of loss-based ABF based on the Gaussian ARCH(1) auxiliary model, plus the predictive accuracy of FBP in the case where the Gaussian ARCH(1) model is used as the assumed predictive model in the generalized updating. Table 6 documents results under correct specification, and Table 7 documents results under one form of misspecification. All results are for the one-step-ahead horizon.

Table 6
Predictive accuracy of loss-based ABF (Panel A) and FBP (Panel B) for one-step-ahead forecasts under the correct specification of the SSM. The Gaussian ARCH(1) model is adopted as the auxiliary model. The true DGP is the SV model in (24)–(26). The rows in Panel A refer to the scoring rule used in generating the summary statistics in the ABC algorithm underlying the loss-based ABF results. The rows in Panel B refer to the scoring rule used in FBP, while the last row in each panel refers to the exact predictive results. The columns in each panel refer to the scoring rule used to compute the out-of-sample average scores. The figures in bold are the largest average scores according to a given out-of-sample measure. The second largest average score is in italics.

Correct model specification							
	Panel A: Loss-based ABF Average out-of-sample score						
	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
Scoring rule							
ABC-LS	*−0.8193*	−0.2959	−0.4844	*−0.4875*	*−0.2985*	*−0.3212*	−3.0605
ABC-CLS10	−0.8215	−0.2965	−0.4855	−0.4883	−0.2991	−0.3214	−3.0728
ABC-CLS20	−0.8200	*−0.2958*	*−0.4842*	−0.4883	−0.2990	−0.3213	−3.0638
ABC-CLS80	−0.8209	−0.2960	**−0.4841**	−0.4888	−0.2994	−0.3215	−3.0642
ABC-CLS90	−0.8215	−0.2960	*−0.4842*	−0.4892	−0.2997	−0.3215	−3.0712
ABC-CRPS	**−0.8191**	**−0.2957**	−0.4844	**−0.4874**	**−0.2984**	**−0.3211**	**−3.0585**
ABC-IS	−0.8194	*−0.2958*	**−0.4841**	−0.4878	−0.2987	*−0.3212*	*−3.0603*
Exact	−0.8191	−0.2957	−0.4842	−0.4876	−0.2986	−0.3211	−3.0594
	Panel B: FBP Average out-of-sample score						
	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
Scoring rule							
FBP-LS	*−0.8990*	−0.3226	−0.5178	*−0.5146*	−0.3200	*−0.3287*	**−3.2972**
FBP-CLS10	−1.1673	*−0.3171*	*−0.5124*	−0.7591	−0.5407	−0.4235	−4.0443
FBP-CLS20	−1.0422	**−0.3170**	**−0.5084**	−0.6453	−0.4341	−0.3710	−3.6861
FBP-CLS80	−1.0273	−0.4176	−0.6327	**−0.5110**	**−0.3175**	−0.3659	−3.6335
FBP-CLS90	−1.1538	−0.5226	−0.7465	−0.5165	*−0.3178*	−0.4171	−3.9913
FBP-CRPS	**−0.8972**	−0.3299	−0.5229	−0.5166	−0.3238	**−0.3255**	−3.3509
FBP-IS	−0.9065	−0.3232	−0.5184	−0.5188	−0.3236	−0.3298	*−3.3073*
Exact	−0.8191	−0.2957	−0.4842	−0.4876	−0.2986	−0.3211	−3.0594

Table 7
Predictive accuracy of loss-based ABF (Panel A) and FBP (Panel B) for one-step-ahead forecasts under misspecification of the SSM. The Gaussian ARCH(1) model is adopted as the auxiliary model. The true DGP is the SV model in (27)–(29) with $\gamma = -5$, whilst the assumed DGP underlying loss-based ABF is the SV model in (24)–(26). The rows in Panel A refer to the scoring rule used in generating the summary statistics in the ABC algorithm underlying the loss-based ABF results. The rows in Panel B refer to the scoring rule used in FBP, while the last row in each panel refers to the exact (but misspecified) predictive results. The columns in each panel refer to the scoring rule used to compute the out-of-sample average scores. The figures in bold are the largest average scores according to a given out-of-sample measure. The second largest average score is in italics.

Model misspecification							
	Panel A: Loss-based ABF Average out-of-sample score						
	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
Scoring rule							
ABC-LS	*−1.3433*	*−0.3591*	−0.6190	−0.4876	−0.2951	*−0.5343*	*−4.6070*
ABC-CLS10	−1.5131	−0.3883	−0.6954	−0.5348	−0.3095	−0.6185	−4.7425
ABC-CLS20	−1.3610	**−0.3545**	*−0.6184*	−0.4975	−0.3005	−0.5411	−4.6149
ABC-CLS80	−1.6676	−0.6114	−0.9617	**−0.4596**	**−0.2742**	−0.5845	−8.0744
ABC-CLS90	−1.8580	−0.6700	−1.0911	*−0.4768*	*−0.2808*	−0.6240	−9.7305
ABC-CRPS	**−1.3359**	−0.3599	**−0.6181**	−0.4911	−0.3027	**−0.5305**	−4.7400
ABC-IS	−1.4926	−0.3786	−0.6873	−0.5290	−0.3053	−0.6125	**−4.3467**
Exact	−1.3343	−0.3618	−0.6199	−0.4882	−0.3003	−0.5304	−4.7357

	Panel B: FBP Average out-of-sample score						
	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
Scoring rule							
FBP-LS	**−1.3691**	−0.3774	−0.6436	−0.4893	−0.3004	*−0.5377*	*−4.8014*
FBP-CLS10	−1.4302	**−0.3678**	*−0.6310*	−0.5621	−0.3764	−0.5490	−5.3863
FBP-CLS20	−1.4781	*−0.3679*	**−0.6289**	−0.6130	−0.4280	−0.5680	−5.7077
FBP-CLS80	−1.9668	−0.8891	−1.2627	**−0.4564**	*−0.2720*	−0.5827	−8.5870
FBP-CLS90	−2.3823	−1.1207	−1.6189	*−0.4620*	**−0.2713**	−0.6174	−10.4630
FBP-CRPS	*−1.3699*	−0.3888	−0.6509	−0.4882	−0.3015	**−0.5347**	−4.9309
FBP-IS	−1.4797	−0.3910	−0.7005	−0.5177	−0.3005	−0.6101	**−4.4381**
Exact	−1.3343	−0.3618	−0.6199	−0.4882	−0.3003	−0.5304	−4.7357

Table 8
Computing times of loss-based ABF (Panel A) versus FBP (Panel B) for one-step-ahead forecasts under misspecification of the SSM. The Gaussian GARCH(1,1) model is adopted as the auxiliary model. The true DGP is the SV model in (27)–(29) with $\gamma = -5$, whilst the assumed DGP underlying loss-based ABF is the SV model in (24)–(26). The rows in Panel A refer to the scoring rule used in generating the summary statistics in the ABC algorithm underlying the loss-based ABF results. The rows in Panel B refer to the scoring rule used in FBP. The columns in Panel A refer to the running times for the reference table in ABC, the distance calculation and the construction of a single predictive under loss-based ABF. The columns in Panel B refer to the running times for posterior estimation via the random walk MH (MCMC) algorithm and construction of a single predictive under FBP. All times are represented in h (Hours), min (Minutes), sec (Seconds) format.

Model misspecification			
	Panel A: Loss-based ABF Running time		
	Reference table	Distance calculation	One predictive construction
Scoring rule			
ABC-LS	00 h, 18 min, 55 s	00 h, 01 min, 02 s	00 h, 04 min, 17 s
ABC-CLS10	00 h, 39 min, 50 s	00 h, 00 min, 55 s	00 h, 04 min, 11 s
ABC-CLS20	00 h, 35 min, 51 s	00 h, 00 min, 49 s	00 h, 04 min, 02 s
ABC-CLS80	00 h, 37 min, 31 s	00 h, 00 min, 50 s	00 h, 03 min, 57 s
ABC-CLS90	00 h, 36 min, 00 s	00 h, 00 min, 51 s	00 h, 03 min, 56 s
ABC-CRPS	00 h, 23 min, 26 s	00 h, 01 min, 05 s	00 h, 04 min, 06 s
ABC-IS	00 h, 40 min, 47 s	00 h, 00 min, 55 s	00 h, 04 min, 00 s

(*continued on next page*)

Table 8 (*continued*).

	Panel B: FBP Running time	
	Posterior density estimation	One predictive construction
Scoring rule		
FBP-LS	00 h, 01 min, 17 s	00 h, 00 min, 01 s
FBP-CLS10	00 h, 01 min, 10 s	00 h, 00 min, 01 s
FBP-CLS20	00 h, 01 min, 06 s	00 h, 00 min, 01 s
FBP-CLS80	00 h, 01 min, 14 s	00 h, 00 min, 01 s
FBP-CLS90	00 h, 01 min, 14 s	00 h, 00 min, 01 s
FBP-CRPS	00 h, 00 min, 42 s	00 h, 00 min, 01 s
FBP-IS	00 h, 01 min, 04 s	00 h, 00 min, 01 s

Table 9
Predictive accuracy of loss-based ABF (Panel A) and FBP (Panel B) for one-step-ahead forecasts under misspecification of the SSM. The Gaussian GARCH(1,1) model is adopted as the auxiliary model. The true DGP is the SV model in (27)–(29) with $\gamma = -2$, whilst the assumed DGP underlying loss-based ABF is the SV model in (24)–(26). The rows in Panel A refer to the scoring rule used in generating the summary statistics in the ABC algorithm underlying the loss-based ABF results. The rows in Panel B refer to the scoring rule used in FBP, while the last row in each panel refers to the exact (but misspecified) predictive results. The columns in each panel refer to the scoring rule used to compute the out-of-sample average scores. The figures in bold are the largest average scores according to a given out-of-sample measure. The second largest average score is in italics.

	Model misspecification						
	Panel A: Loss-based ABF Average out-of-sample score						
	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
Scoring rule							
ABC-LS	*−1.3407*	*−0.3448*	**−0.5913**	−0.5277	*−0.3145*	*−0.5333*	−4.4235
ABC-CLS10	−1.4203	−0.3454	−0.6061	−0.5597	−0.3271	−0.5669	*−4.3370*
ABC-CLS20	−1.3647	**−0.3436**	−0.5935	−0.5373	−0.3178	−0.5428	−4.3673
ABC-CLS80	−1.5635	−0.4991	−0.7918	**−0.5255**	**−0.3141**	−0.5679	−6.4956
ABC-CLS90	−1.7361	−0.5855	−0.9268	−0.5321	−0.3158	−0.5950	−7.8254
ABC-CRPS	**−1.3382**	−0.3464	*−0.5930*	*−0.5264*	−0.3149	**−0.5324**	−4.4644
ABC-IS	−1.3806	−0.3450	−0.6000	−0.5417	−0.3188	−0.5521	**−4.3198**
Exact	−1.3349	−0.3449	−0.5915	−0.5251	−0.3137	−0.5320	−4.4602
	Panel B: FBP Average out-of-sample score						
	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
Scoring rule							
FBP-LS	**−1.3399**	−0.3495	−0.5978	−0.5220	−0.3105	*−0.5338*	*−4.4851*
FBP-CLS10	−1.3461	**−0.3460**	*−0.5964*	−0.5279	−0.3163	−0.5354	−4.5374
FBP-CLS20	−1.3581	*−0.3465*	**−0.5942**	−0.5446	−0.3325	−0.5365	−4.7246
FBP-CLS80	−1.4348	−0.4429	−0.7002	**−0.5156**	*−0.3048*	−0.5445	−5.2840
FBP-CLS90	−1.5035	−0.4928	−0.7643	*−0.5169*	**−0.3042**	−0.5558	−5.7729
FBP-CRPS	*−1.3407*	−0.3520	−0.5991	−0.5231	−0.3120	**−0.5331**	−4.5328
FBP-IS	−1.3680	−0.3488	−0.6092	−0.5289	−0.3115	−0.5508	**−4.3809**
Exact	−1.3349	−0.3449	−0.5915	−0.5251	−0.3137	−0.5320	−4.4602

(ii) The computing time for both focused methods, loss-based ABF and FBP, under one form of misspecification and for the one-step-ahead horizon. This information (documented in Table 8) supplements the discussion in Footnote 3.

(iii) The predictive accuracy of loss-based ABF and FBP for one-step-ahead forecasts under additional model misspecification scenarios (Tables 9 and 10).

(iv) The predictive accuracy of loss-based ABF and FBP for longer forecast horizons. Table 11 documents results for the five-step-ahead horizon, and Table 12 documents results for the ten-step-ahead horizon, each under one form of misspecification.

Table 10
Predictive accuracy of loss-based ABF (Panel A) and FBP (Panel B) for one-step-ahead forecasts under misspecification of the SSM. The Gaussian GARCH(1,1) model is adopted as the auxiliary model. The true DGP is the SV model in (27)–(29) with $\gamma = -1$, whilst the assumed DGP underlying loss-based ABF is the SV model in (24)–(26). The rows in Panel A refer to the scoring rule used in generating the summary statistics in the ABC algorithm underlying the loss-based ABF results. The rows in Panel B refer to the scoring rule used in FBP, while the last row in each panel refers to the exact (but misspecified) predictive results. The columns in each panel refer to the scoring rule used to compute the out-of-sample average scores. The figures in bold are the largest average scores according to a given out-of-sample measure. The second largest average score is in italics.

Model misspecification

Panel A: Loss-based ABF
Average out-of-sample score

Scoring rule	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
ABC-LS	**−1.3445**	−0.3318	**−0.5679**	**−0.5500**	−0.3257	*−0.5345*	*−4.3198*
ABC-CLS10	−1.4782	*−0.3317*	−0.5797	−0.6288	−0.3716	−0.5805	−4.7755
ABC-CLS20	−1.3971	−0.3342	−0.5722	−0.5803	−0.3431	−0.5520	−4.3806
ABC-CLS80	−1.4428	−0.3878	−0.6477	−0.5493	*−0.3248*	−0.5577	−5.0186
ABC-CLS90	−1.5524	−0.4372	−0.7261	−0.5560	**−0.3243**	−0.5847	−5.8805
ABC-CRPS	*−1.3462*	−0.3333	*−0.5693*	*−0.5524*	−0.3285	**−0.5336**	−4.4792
ABC-IS	−1.3516	**−0.3315**	**−0.5679**	−0.5534	−0.3274	−0.5382	**−4.2774**
Exact	−1.3396	−0.3306	−0.5666	−0.5478	−0.3241	−0.5342	−4.3276

Panel B: FBP
Average out-of-sample score

Scoring rule	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
FBP-LS	**−1.3389**	−0.3315	*−0.5678*	−0.5459	−0.3221	−0.5349	*−4.3831*
FBP-CLS10	−1.3453	**−0.3307**	−0.5690	−0.5488	−0.3241	−0.5383	−4.3720
FBP-CLS20	*−1.3391*	−0.3311	**−0.5676**	−0.5462	−0.3223	−0.5349	−4.3890
FBP-CLS80	−1.3537	−0.3469	−0.5840	**−0.5447**	*−0.3208*	−0.5373	−4.4943
FBP-CLS90	−1.3789	−0.3653	−0.6069	*−0.5458*	**−0.3201**	**−0.5437**	−4.6686
FBP-CRPS	−1.3400	−0.3323	−0.5683	−0.5470	−0.3233	*−0.5347*	−4.4186
FBP-IS	−1.3433	*−0.3308*	−0.5693	−0.5463	−0.3220	−0.5375	**−4.3434**
Exact	−1.3396	−0.3306	−0.5666	−0.5478	−0.3241	−0.5342	−4.3276

Table 11
Predictive accuracy of loss-based ABF (Panel A) and FBP (Panel B) for five-step-ahead forecasts under misspecification of the SSM. The Gaussian GARCH(1,1) model is adopted as the auxiliary model. The true DGP is the SV model in (27)–(29) with $\gamma = -5$, whilst the assumed DGP underlying loss-based ABF is the SV model in (24)–(26). The rows in Panel A refer to the scoring rule used in generating the summary statistics in the ABC algorithm underlying the loss-based ABF results. The rows in Panel B refer to the scoring rule used in FBP, while the last row in each panel refers to the exact (but misspecified) predictive results. The columns in each panel refer to the scoring rule used to compute the out-of-sample average scores. The figures in bold are the largest average scores according to a given out-of-sample measure. The second largest average score is in italics.

Model misspecification

Panel A: Loss-based ABF
Average out-of-sample score

Scoring rule	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
ABC-LS	*−1.3742*	−0.3716	**−0.6316**	−0.4998	−0.3053	*−0.5384*	−4.8524
ABC-CLS10	−1.4497	*−0.3681*	−0.6435	−0.5267	−0.3140	−0.5680	*−4.7737*
ABC-CLS20	−1.4091	**−0.3679**	*−0.6336*	−0.5202	−0.3202	−0.5485	−5.0914
ABC-CLS80	−2.2578	−0.8871	−1.4099	**−0.4894**	*−0.3028*	−0.6153	−11.3280
ABC-CLS90	−2.6141	−0.9675	−1.6080	*−0.4897*	**−0.2962**	−0.6583	−13.5940
ABC-CRPS	**−1.3646**	−0.3750	−0.6347	−0.4960	−0.3067	**−0.5357**	−4.9569
ABC-IS	−1.5298	−0.3758	−0.6746	−0.5530	−0.3193	−0.6095	**−4.4957**
Exact	−1.3620	−0.3737	−0.6332	−0.4963	−0.3072	−0.5352	−4.9536

(*continued on next page*)

Table 11 (*continued*).

	Panel B: FBP Average out-of-sample score						
	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
Scoring rule							
FBP-LS	**−1.3772**	−0.3811	−0.6471	−0.4925	−0.3045	*−0.5389*	*−4.8918*
FBP-CLS10	−1.4198	*−0.3744*	*−0.6431*	−0.5321	−0.3442	−0.5476	−5.2016
FBP-CLS20	−1.4580	**−0.3711**	**−0.6335**	−0.5829	−0.3978	−0.5570	−5.5652
FBP-CLS80	−1.9282	−0.8633	−1.2277	**−0.4583**	**−0.2730**	−0.5850	−8.5684
FBP-CLS90	−2.2912	−1.0555	−1.5326	*−0.4725*	*−0.2761*	−0.6254	−10.4330
FBP-CRPS	*−1.3773*	−0.3883	−0.6511	−0.4928	−0.3065	**−0.5366**	−4.9966
FBP-IS	−1.4776	−0.3918	−0.6959	−0.5175	−0.3018	−0.6020	**−4.5027**
Exact	−1.3620	−0.3737	−0.6332	−0.4963	−0.3072	−0.5352	−4.9536

Table 12
Predictive accuracy of loss-based ABF (Panel A) and FBP (Panel B) for ten-step-ahead forecasts under misspecification of the SSM. The Gaussian GARCH(1,1) model is adopted as the auxiliary model. The true DGP is the SV model in (27)–(29) with $\gamma = -5$, whilst the assumed DGP underlying loss-based ABF is the SV model in (24)–(26). The rows in Panel A refer to the scoring rule used in generating the summary statistics in the ABC algorithm underlying the loss-based ABF results. The rows in Panel B refer to the scoring rule used in FBP, while the last row in each panel refers to the exact (but misspecified) predictive results. The columns in each panel refer to the scoring rule used to compute the out-of-sample average scores. The figures in bold are the largest average scores according to a given out-of-sample measure. The second largest average score is in italics.

	Model misspecification						
	Panel A: Loss-based ABF Average out-of-sample score						
	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
Scoring rule							
ABC-LS	*−1.3892*	−0.3773	**−0.6381**	−0.5048	*−0.3087*	*−0.5409*	−4.9532
ABC-CLS10	−1.4720	**−0.3729**	−0.6481	−0.5361	−0.3197	−0.5717	*−4.8791*
ABC-CLS20	−1.4288	*−0.3737*	*−0.6399*	−0.5291	−0.3273	−0.5517	−5.3026
ABC-CLS80	−2.3613	−0.9469	−1.4934	*−0.4987*	−0.3113	−0.6176	−11.6380
ABC-CLS90	−2.7462	−1.0329	−1.7079	**−0.4982**	**−0.3045**	−0.6608	−13.8790
ABC-CRPS	**−1.3770**	−0.3800	−0.6405	−0.4998	−0.3098	**−0.5378**	−5.0616
ABC-IS	−1.5528	−0.3805	−0.6791	−0.5649	−0.3266	−0.6130	**−4.6521**
Exact	−1.3752	−0.3786	−0.6390	−0.5007	−0.3108	−0.5374	−5.0552
	Panel B: FBP Average out-of-sample score						
	LS	CLS10	CLS20	CLS80	CLS90	CRPS	IS
Scoring rule							
FBP-LS	*−1.3876*	*−0.3851*	*−0.6519*	−0.4964	*−0.3082*	*−0.5404*	*−4.9588*
FBP-CLS10	−1.4779	−0.3881	−0.6574	−0.5640	−0.3756	−0.5611	−5.6479
FBP-CLS20	−1.5336	**−0.3847**	**−0.6483**	−0.6248	−0.4388	−0.5775	−6.1194
FBP-CLS80	−2.3341	−1.0913	−1.5689	**−0.4705**	−0.2859	−0.6014	−10.2470
FBP-CLS90	−2.8345	−1.2559	−1.9411	*−0.4755*	**−0.2813**	−0.6457	−12.6090
FBP-CRPS	**−1.3871**	−0.3939	−0.6572	−0.4949	−0.3084	**−0.5379**	−5.0616
FBP-IS	−1.4995	−0.3979	−0.7028	−0.5281	−0.3083	−0.6051	**−4.6194**
Exact	−1.3752	−0.3786	−0.6390	−0.5007	−0.3108	−0.5374	−5.0552

References

Andrieu, C., Doucet, A., & Holenstein, R. (2011). Particle Markov chain Monte Carlo. *Journal of the Royal Statistical Society. Series B. Statistical Methodology*, *72*(2), 269–342, With discussion.

Biau, G., Cérou, F., & Guyader, A. (2015). New insights into approximate Bayesian computation. In *Annales de l'IHP Probabilités et statistiques*: *vol. 51*, (pp. 376–403).

Bissiri, P. G., Holmes, C. C., & Walker, S. G. (2016). A general framework for updating belief distributions. *Journal of the Royal Statistical Society. Series B. Statistical Methodology*, *78*(5), 1103–1130.

Blei, D. M., Kucukelbir, A., & McAuliffe, J. D. (2017). Variational inference: A review for statisticians. *Journal of the American Statistical Association*, *112*(518), 859–877.

Blum, M. G. (2010). Approximate Bayesian computation: A nonparametric perspective. *Journal of the American Statistical Association*, *105*(491), 1178–1187.

Carr, P., & Wu, L. (2003). The finite moment log stable process and option pricing. *The Journal of Finance*, *58*(2), 753–777.

Carter, C. K., & Kohn, R. (1994). On Gibbs sampling for state space models. *Biometrika*, *81*(3), 541–553.

Chakraborty, A., Nott, D. J., Drovandi, C. C., Frazier, D. T., & Sisson, S. A. (2023). Modularized Bayesian analyses and cutting feedback in likelihood-free inference. *Statistics and Computing*, *33*(1), 33.

Chan, J. C.-C., & Jeliazkov, I. (2009). MCMC estimation of restricted covariance matrices. *Journal of Computational and Graphical Statistics*, *18*(2), 457–480.

Chan, J. C., & Yu, X. (2022). Fast and accurate variational inference for large Bayesian VARs with stochastic volatility. *Journal of Economic Dynamics & Control*, *143*, Article 104505.

Cornuet, J.-M., Santos, F., Beaumont, M. A., Robert, C. P., Marin, J.-M., Balding, D. J., Guillemaud, T., & Estoup, A. (2008). Inferring population history with DIY ABC: A user-friendly approach to approximate Bayesian computation. *Bioinformatics*, *24*(23), 2713–2719.

Creel, M., & Kristensen, D. (2015). ABC of SV: Limited information likelihood inference in stochastic volatility jump-diffusion models. *Journal of Empirical Finance*, *31*, 85–108.

Dean, T. A., Singh, S. S., Jasra, A., & Peters, G. W. (2014). Parameter estimation for hidden Markov models with intractable likelihoods. *Scandinavian Journal of Statistics*, *41*(4), 970–987.

Diks, C., Panchenko, V., & Van Dijk, D. (2011). Likelihood-based scoring rules for comparing density forecasts in tails. *Journal of Econometrics*, *163*(2), 215–230.

Drovandi, C. C., Pettitt, A. N., & Faddy, M. J. (2011). Approximate Bayesian computation using indirect inference. *Journal of the Royal Statistical Society. Series C. Applied Statistics*, *60*(3), 317–337.

Drovandi, C. C., Pettitt, A. N., & Lee, A. (2015). Bayesian indirect inference using a parametric auxiliary model. *Statistical Science*, *30*(1), 72–95.

Fearnhead, P. (2011). Bayesian inference for time series state space models. In S. Brooks, A. Gelman, G. Jones, & X. Meng (Eds.), *Handbook of markov chain Monte Carlo* (pp. 513–530). Taylor & Francis.

Fearnhead, P., & Prangle, D. (2012). Constructing summary statistics for approximate Bayesian computation: Semi-automatic approximate Bayesian computation. *Journal of the Royal Statistical Society. Series B. Statistical Methodology*, *74*(3), 419–474.

Flury, T., & Shephard, N. (2011). Bayesian inference based only on a simulated likelihood. *Econometric Theory*, *27*, 933–956.

Frazier, D. T., Loaiza-Maya, R., & Martin, G. M. (2023). Variational Bayes in state space models: Inferential and predictive accuracy. *Journal of Computational and Graphical Statistics*, *32*(3), 793–804.

Frazier, D. T., Loaiza-Maya, R., Martin, G. M., & Koo, B. (2024). Loss-based variational Bayes prediction. *Journal of Computational and Graphical Statistics*, (to appear).

Frazier, D. T., Maneesoonthorn, W., Martin, G. M., & McCabe, B. P. (2019). Approximate Bayesian forecasting. *International Journal of Forecasting*, *35*(2), 521–539.

Frazier, D. T., Martin, G. M., Robert, C. P., & Rousseau, J. (2018). Asymptotic properties of approximate Bayesian computation. *Biometrika*, *105*(3), 593–607.

Frazier, D. T., Robert, C. P., & Rousseau, J. (2020). Model misspecification in approximate Bayesian computation: Consequences and diagnostics. *Journal of the Royal Statistical Society. Series B. Statistical Methodology*, *82*(2), 421–444.

Frühwirth-Schnatter, S. (1994). Data augmentation and dynamic linear models. *Journal of Time Series Analysis*, *15*(2), 183–202.

Giordani, P., Pitt, M., & Kohn, R. (2011). Bayesian inference for time series state space models. In J. Geweke, G. Koop, & H. Van Dijk (Eds.), *The Oxford Handbook of Bayesian Econometrics* (pp. 61–124). OUP.

Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, *102*(477), 359–378.

Gordon, N. J., Salmond, D. J., & Smith, A. F. (1993). Novel approach to nonlinear/non-Gaussian Bayesian state estimation. In *IEE proceedings F (radar and signal processing): vol. 140*, (2), (pp. 107–113). IET.

Jacquier, E., & Polson, N. (2011). Bayesian methods in finance. *The Oxford Handbook of Bayesian Econometrics*, 439–512, OUP. Eds. Geweke, J., Koop, G. and Van Dijk, H..

Jacquier, R., Polson, N. G., & Rossi, P. E. (1994). Bayesian analysis of stochastic volatility models. *Journal of Business & Economic Statistics*, *12*(4), 371–389, With discussion.

Jiang, W., & Tanner, M. A. (2008). Gibbs posterior for variable selection in high-dimensional classification and data mining. *The Annals of Statistics*, *36*(5), 2207–2231.

Joyce, P., & Marjoram, P. (2008). Approximately sufficient statistics and Bayesian computation. *Statistical Applications in Genetics and Molecular Biology*, *7*(1).

Kim, S., Shephard, N., & Chib, S. (1998). Stochastic volatility: Likelihood inference and comparison with ARCH models. *Review of Economic Studies*, *65*(3), 361–393.

Koop, G., & Korobilis, D. (2023). Bayesian dynamic variable selection in high dimensions. *International Economic Review*, *64*(3), 1047–1074.

Lacoste-Julien, S., Huszár, F., & Ghahramani, Z. (2011). Approximate inference for the loss-calibrated Bayesian. In *Proceedings of the fourteenth international conference on artificial intelligence and statistics* (pp. 416–424). JMLR Workshop and Conference Proceedings.

Loaiza-Maya, R., Martin, G. M., & Frazier, D. T. (2021). Focused Bayesian prediction. *Journal of Applied Econometrics*, *36*(5), 517–543.

Loaiza-Maya, R., Smith, M. S., Nott, D. J., & Danaher, P. J. (2022). Fast and accurate variational inference for models with many latent variables. *Journal of Econometrics*, *230*(2), 339–362.

Lombardi, M. J., & Calzolari, G. (2009). Indirect estimation of α-stable stochastic volatility models. *Computational Statistics & Data Analysis*, *53*(6), 2298–2308.

Martin, G. M., Frazier, D. T., & Robert, C. P. (2024). Approximating Bayes in the 21st century. *Statistical Science*, *39*(1), 20–45.

Martin, G. M., Loaiza-Maya, R., Maneesoonthorn, W., Frazier, D. T., & Ramírez-Hassan, A. (2022). Optimal probabilistic forecasts: When do they work? *International Journal of Forecasting*, *38*(1), 384–406.

Martin, G. M., McCabe, B. P., Frazier, D. T., Maneesoonthorn, W., & Robert, C. P. (2019). Auxiliary likelihood-based approximate Bayesian computation in state space models. *Journal of Computational and Graphical Statistics*, *28*(3), 508–522.

Nott, D. J., Drovandi, C., & Frazier, D. T. (2023). Bayesian inference for misspecified generative models. *Annual Review of Statistics and Its Application*, *11*.

Peters, G. W., Sisson, S. A., & Fan, Y. (2012). Likelihood-free Bayesian inference for α-stable models. *Computational Statistics & Data Analysis*, *56*(11), 3743–3756.

Pritchard, J. K., Seielstad, M. T., Perez-Lezaun, A., & Feldman, M. W. (1999). Population growth of human Y chromosomes: A study of Y chromosome microsatellites.. *Molecular Biology and Evolution*, *16*(12), 1791–1798.

Quiroz, M., Nott, D. J., & Kohn, R. (2023). Gaussian variational approximations for high-dimensional state space models. *Bayesian Analysis*, *18*(3), 989–1016.

Strickland, C. M., Forbes, C. S., & Martin, G. M. (2006). Bayesian analysis of the stochastic conditional duration model. *Computational Statistics & Data Analysis*, *50*(9), 2247–2267.

Stroud, J. R., Müller, P., & Polson, N. G. (2003). Nonlinear state-space models with state-dependent variances. *Journal of the American Statistical Association*, *98*(462), 377–386.

Tavaré, S., Balding, D. J., Griffiths, R. C., & Donnelly, P. (1997). Inferring coalescence times from DNA sequence data. *Genetics*, *145*(2), 505–518.

Tran, M.-N., Nott, D. J., & Kohn, R. (2017). Variational Bayes with intractable likelihood. *Journal of Computational and Graphical Statistics*, *26*(4), 873–882.

Wegmann, D., Leuenberger, C., & Excoffier, L. (2009). Efficient approximate Bayesian computation coupled with Markov chain Monte Carlo without likelihood. *Genetics*, *182*(4), 1207–1218.

Zhang, T. (2006). Information-theoretic upper and lower bounds for statistical estimation. *Institute of Electrical and Electronics Engineers. Transactions on Information Theory*, *52*(4), 1307–1321.

Zhang, C., Bütepage, J., Kjellström, H., & Mandt, S. (2018). Advances in variational inference. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *41*(8), 2008–2026.